

Coherence time in biological oscillator assemblies bounds the rate of state registration

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February 2026

Abstract

~~Cognitive and biological processing depends on coherent oscillations across distributed assemblies. Yet the~~ The physical principles determining how fast ~~these distributed~~ neural assemblies can synchronize—and thus how fast ~~neural processes associated with the neural processes underlying~~ cognition can proceed—remain incompletely formalized. We derive a ~~quantitative framework showing that coherence time in~~ coherence time bound for coupled oscillator networks: the minimum interval between irreversible state registrations (“commits”) scales exponentially with coordination depth M . For M semi-independent modules requiring phase alignment ~~within tolerance ϵ~~ at coherence r , the ~~minimum interval between irreversible state registrations (“commits”)~~ commit rate is dominated by rare-event first-passage time ~~into low-measure alignment sets~~ on the configuration ~~manifold. This produces~~ torus, producing a fundamental speed-flexibility ~~trade-off: increasing coordination depth~~ trade-off—increasing M expands combinatorial flexibility but slows commits exponentially; increasing ~~coherence~~ r accelerates synchronization but restricts dynamics to low-dimensional attractors. We identify the

pre-commit *phase delta regime*—in which structured inter-module phase relationships bias downstream outcomes without discrete state registration—as a computationally productive mode operating below the Landauer threshold, distinct from phase synchrony as conventionally understood in the binding literature. We situate this coordination limit within a hierarchy of physical ~~bounds—quantum speed limits~~, bounds including quantum speed limits and Landauer-Margolus-Levitin thermodynamic constraints, ~~and energy-time uncertainty relations—showing that~~ showing that coordination dominates by eleven orders of magnitude in biological neural systems, ~~these quantum and thermodynamic bounds are typically non-binding; the dominant constraint is the time required for distributed oscillatory assemblies to achieve sufficient phase coherence. While neural oscillators provide the most empirically accessible measurements, the framework applies to any biological system whose function requires coordinated phase relations and sparse commitment events, from chromosomal segregation to circadian networks. We provide parameter estimation protocols and testable predictions including alpha entrainment effects on temporal acuity and~~. We validate the scaling law in Kuramoto simulations ($R^2 = 0.97$ for modular networks) and show that the binding window calculation reproduces the empirical 30–50 ms range from independently constrained parameters. The framework generates testable predictions: alpha entrainment via tACS should shift binding windows linearly with frequency; the ratio $D_{\text{eff}}/\Gamma_{\text{commit}}$ predicts subjective temporal scaling, yielding qualitatively distinct predictions across four pharmacological classes from a two-parameter structure; and subjective time dilation under acute stress is explained as a dual-loop ~~dissociations under arousal~~ dissociation between high- M perceptual and low- M motor commit architectures.

Keywords: biological oscillations, coherence time, coordination depth, temporal resolution, quantum speed limits, Landauer bound, coupled oscillators

1 Introduction

1.1 Processing speed in biological oscillator assemblies

A fundamental insight from systems neuroscience is that cognition emerges from coherent oscillations across neural assemblies, not merely from individual spike rates. Working memory maintenance involves theta-gamma phase-amplitude coupling organized in discrete oscillatory bursts [1, 6, 2]. Attention selectively enhances inter-areal synchronization in gamma band [4, 7]. Perceptual binding depends on transient phase alignment across visual cortices [5, 8]. Long-range communication occurs preferentially during coherent states [9]. The “communication through coherence” framework [10] has become central to understanding neural computation.

Yet despite extensive empirical characterization, the physical principles governing *how fast* distributed assemblies can achieve coherence—and thus how quickly the neural processes underlying cognitive operations can proceed—remain incompletely formalized. Why does perceptual binding require 30–50 ms rather than 3 ms or 300 ms? Why do larger assemblies integrating more information process more slowly? What determines the relationship between oscillation frequency and temporal acuity?

1.2 Coherence time as a physical bound on processing rate

Recent work argued that living intelligence is an emergent property of high-dimensional, self-sustaining dynamical systems that maintain coherence against decoherence [11], building on analyses of falsifiability limits in adaptive biological systems [41] and timing inaccessibility in sub-Landauer computation [42]. The present manuscript addresses the complementary question: given such a coherent high-dimensional system, what sets the fastest timescale on which it can register change and update its state?

We formalize this as the minimum interval between irreversible “commit” events—transitions in which distributed dynamics become readable to downstream mechanisms, consistent with

the emerging view that perception operates in discrete cycles [45]. This commit interval is bounded by a coordination term (coherence time as a first-passage problem on a product manifold) together with quantum, noise, and power limits.

Operational definition of state registration. A registration (or “commit”) is an event that (i) requires multi-module coordination—simultaneous phase alignment across semi-independent neural assemblies, (ii) produces a stable macrostate with measurable dwell time, and (iii) is detectable as a discrete update in behaviour, downstream neural population state, or EEG/MEG decoding accuracy. The framework addresses the temporal resolution of such events; it does not claim that all neural computation reduces to discrete registrations.

Operational definition of commits in neural systems. A commit is operationally detected when distributed oscillatory dynamics cross a threshold that enables downstream readout. Empirical signatures include: (i) phase-locking index exceeding threshold across specified modules for a minimum duration, (ii) onset of stable cross-frequency coupling (e.g., theta-gamma PAC), (iii) classifier-detectable state transition in downstream neural populations. We do not claim all neural computation reduces to discrete commits; rather, we claim that the *fastest update of downstream state* is bounded by the time required for multi-module phase alignment. The framework addresses temporal resolution limits, not the full richness of neural dynamics.

Definition 1 (Commit event). *Given M modules with aggregate phases $\psi_1(t), \dots, \psi_M(t)$ and within-module coherences $r_{\text{intra},m}(t)$, a **commit** occurs at time t^* if:*

1. *All modules are internally coherent: $r_{\text{intra},m}(t^*) \geq r_{\text{thr}}$ for all m .*
2. *Phases are mutually aligned: $\max_m |\psi_m(t^*) - \bar{\psi}(t^*)| \leq \varepsilon/2$, where $\bar{\psi}$ is the circular mean.*
3. *The alignment persists for dwell time T_{dwell} .*

The commit is defined relative to a downstream readout mechanism; different readouts may have different $(r_{\text{thr}}, \varepsilon, T_{\text{dwell}})$ parameters.

Empirical proxies for commit detection: PLI threshold crossing (sustained $r_{\text{inter}} > r_{\text{thresh}}$),
PAC onset, or classifier-detectable state change. These are measurement definitions; different
tasks may require different readout criteria.

What commit events are (and aren't). This framework does **not** claim cognition is discrete or that neural activity has sharp “commits.” It claims the **fastest reliable downstream readout** of multi-module coordination is lower-bounded by the timescale of phase alignment.

Empirical proxies for commit detection:

- ~~Phase-locking index (PLI) threshold crossing: sustained $r_{\text{inter}} > r_{\text{thresh}}$ for dwell time t_{dwell}~~
- **Phase-amplitude coupling (PAC) onset:** stable cross-frequency coupling emerges
- **Classifier-detectable state change:** e.g., decoding accuracy crosses threshold

These are **measurement definitions**, not ontological claims. Different tasks may require different readout criteria.

Minimal empirical pipeline. To estimate coherence time from neural data:

1. Identify modules via connectivity/functional clustering $\rightarrow$ count M
2. Extract module-level phases $\psi_m(t)$ via Hilbert transform or band-pass filtering
3. Compute intra- and inter-module coherence: $r_{\text{intra}}, r_{\text{inter}}$
4. Define alignment threshold ε from task requirements (e.g., decoding accuracy)
5. Estimate attempt rate λ_{attempt} from decorrelation time of $r_{\text{inter}}(t)$
6. Measure first-passage times to sustained alignment $\rightarrow$ fit Eq. 4

Definition (~~Ontological~~ Physical dimensionality). Throughout this work, “dimensionality” refers to ~~a physical property of the system, not a mathematical measure.~~
~~Ontological~~ the effective degrees of freedom of accessible dynamics—not geometric embedding dimension. Physical dimensionality is a quantitative measure of effective connectivity: how many independent directions of coordinated variation are accessible to a system given its

interaction structure. High-dimensional states correspond to weakly constrained dynamics in which many components vary semi-independently; low-dimensional states correspond to strongly coupled dynamics restricted to a small number of collective modes. Dimensionality in this sense is determined not by component count, but by the rank of the system’s interaction structure. Different measurement schemes (participation ratios, manifold dimension, information-geometric metrics) ~~provide projections of this underlying physical property, but do not constitute it~~ estimate this property from data [47].

~~This ontological~~ Physical dimensionality is state-dependent: the same ~~physical~~ system may occupy high- or low-dimensional regimes depending on coupling strength and task demands. Coherence time depends on ~~ontological dimensionality—specifically,~~ how many semi-independent constraints must be simultaneously satisfied for a commit to occur—not on any particular estimator.

Operationalizing dimensionality. ~~While ontological dimensionality is a physical property, it~~ Physical dimensionality must be estimated from measurable quantities. For the coherence time framework, we operationalize effective dimensionality as:

1. **Coordination depth** M : Count of semi-independent modules (functional clusters) whose order parameters must jointly exceed threshold. Estimated via community detection on band-limited coherence networks or anatomical parcellation.
2. **Effective independence** α : Fraction of potential constraint dimensions that are actually independent. Estimated via:
 - Participation ratio of module-phase covariance (Eq. 5),
 - or empirical scaling of τ with M across task conditions.
3. **Effective constraint dimension**: $d_{\text{eff}} = \alpha(M - 1)$ represents the number of semi-independent coordination requirements for a commit.

These are *estimators* of the underlying physical property. Future work may employ other proxies (information-geometric metrics, topological data analysis, dynamical systems dimen-

sion estimates), but the coherence time itself depends on the physical constraint structure, not the measurement method.

1.3 Structure of this work

We first derive the coherence time formula from principles of phase synchronization in coupled oscillator networks (§2.1), then ~~embed it in a unified temporal resolution bound combining quantum, noise, and power constraints~~ characterize the speed-flexibility trade-off (§2.2). Section 2.3 identifies the pre-commit phase delta regime as a computationally productive mode and connects it to sub-Landauer operation. Section 2.4 situates our framework ~~explicitly within the quantum and thermodynamic limits of~~ within the hierarchy of physical limits on computation, addressing the Landauer bound, Margolus-Levitin quantum speed limit, and energy-time uncertainty relation. Section 2.5 provides parameter estimation protocols. Section 3 applies the framework to neural systems, demonstrating quantitative agreement with perceptual binding windows, tachypsychia dissociations, and metabolic scaling. Section 4 discusses ~~the speed-flexibility trade-off, delay embedding as a preservation mechanism, generalization beyond neural systems,~~ generalization, the relationship between frequency and dimensionality, limitations, and testable predictions.

2 Theory and Methods

2.1 The unified temporal resolution bound

We model biological temporal processing as a sequence of *commits*—thermodynamically irreversible events that register high-dimensional internal state as low-dimensional output. A commit requires: (1) dimensional collapse from distributed state (D_{eff} dimensions) to registered outcome (few dimensions), (2) dissipation of at least $k_{\text{B}}T \ln 2$ per bit (Landauer bound), and (3) creation of a persistent measurement influencing future dynamics.

The minimum time between commits is bounded below by four physical constraints:

$$\tau_{\text{eff}} \gtrsim \max[\tau_{\text{QSL}}, \tau_{\text{SNR}}, \tau_{\text{coh}}, \tau_{\text{power}}] \quad (1)$$

This is a **lower bound**: the effective commit interval must be at least as long as the slowest constraint. The max operation is appropriate when these constraints act on largely independent subsystems that can be satisfied in parallel (e.g., quantum state evolution, signal detection, and phase coordination proceed concurrently). If constraints were strictly sequential (e.g., signal detection *then* coordination *then* dissipation), a sum would be more appropriate. In biological neural systems, the constraints are largely parallel, and the coherence time dominates by orders of magnitude, so the distinction has minimal practical impact.

Epistemic status of each term. ~~We distinguish: **Hard physical bounds** (τ_{QSL} ; and the Landauer energy-per-bit):~~ ~~These follow from fundamental physics and cannot be violated regardless of engineering or biological design. **Model-based lower bounds** (are hard physical bounds; τ_{SNR}):~~ ~~These assume specific noise models (AWGN, matched filtering); different noise structures or detection strategies could yield different bounds, though the order of magnitude is robust. **Coordination-time estimates** (and τ_{power} are model-dependent; τ_{coh}):~~ ~~These are is an asymptotic rare-event estimates whose prefactor depends on mixing assumptions and correlation structure. The estimate whose scaling exponent ($M - 1$) is robust ; the prefactor requires empirical calibration. **Power constraints** (τ_{power}):~~ ~~Model-dependent, assuming a specific relationship between dimensional collapse and energy cost. but whose prefactor requires calibration.~~ The key claim is not that Eq. 1 is a theorem, but that τ_{coh} dominates by orders of magnitude ~~in biological systems~~, making the precise form of other constraints largely irrelevant ~~for practical predictions~~.

The four terms are:

Quantum speed limits (τ_{QSL}). The Mandelstam-Tamm and Margolus-Levitin bounds [14, 15] establish

$$\tau_{\text{QSL}} = \max \left[\frac{\hbar}{2\Delta E}, \frac{\pi\hbar}{2\langle E \rangle} \right]$$

For biological temperatures ($\Delta E \sim k_{\text{B}}T$), $\tau_{\text{QSL}} \sim 10^{-13}$ s.

Signal-to-noise limit (τ_{SNR}). Matched-filter detection in additive white Gaussian noise (two-sided power spectral density N_0 , units: power/Hz) over bandwidth B achieves $\text{SNR}(T) = 2P_s T / N_0$ where P_s is signal power and T is integration time [20]. Reaching detection threshold $\rho_{\text{min}} \sim 5$ –10 requires

$$\tau_{\text{SNR}} = \frac{\rho_{\text{min}} N_0}{2P_s}$$

Power limit (τ_{power}). Dimensional collapse by $\Delta\mathcal{D}$ at dimensional chemical potential $\lambda_{\mathcal{D}} \sim k_{\text{B}}T_{\text{eff}}$ costs energy $\sim \lambda_{\mathcal{D}}\Delta\mathcal{D}$. At sustained power P , commit rate is bounded by

$$\tau_{\text{power}} = \frac{\lambda_{\mathcal{D}}\Delta\mathcal{D}}{P}$$

For cortical visual binding: $\Delta\mathcal{D} \sim 50$ (participation ratio of visual cortex assemblies collapsing to a ~ 5 -bit readout), $\lambda_{\mathcal{D}} \geq k_{\text{B}}T \ln 2 \approx 3 \times 10^{-21}$ J (Landauer floor), and $P \sim 10^{-4}$ W (power budget of a $\sim 10^4$ -neuron cortical assembly at ~ 10 nW/neuron). This yields $\tau_{\text{power}} \sim 10^{-15}$ s at the Landauer minimum. Biological neurons are energetically inefficient: ion-channel computation dissipates $\sim 10^7$ times the Landauer floor per degree of freedom [12]. Even at this biophysical scale, $\tau_{\text{power}} \sim 10^{-8}$ s—still six orders of magnitude below τ_{coh} . The power limit is therefore non-binding in the neural regime.

Coherence time (τ_{coh}). We model commits as threshold crossings requiring M semi-independent modules to achieve simultaneous phase alignment within tolerance ε , building on the theory of coupled oscillators [35, 37]. Let r denote the **inter-module coherence** $r_{\text{inter}} = |M^{-1} \sum_m e^{i\psi_m}|$ where ψ_m is the aggregate phase of module m . (We reserve r_{global}

for the all-oscillator order parameter, which serves as a diagnostic but does not appear in Eq. 4.) The von Mises concentration $\kappa(r)$ satisfies $r = I_1(\kappa)/I_0(\kappa)$ [22].

Why von Mises? Given circular symmetry and a fixed mean resultant length r , the maximum-entropy distribution on the circle is von Mises [22]. This makes our modeling choice the canonical least-informative closure: $r \rightarrow \kappa \rightarrow p_1$ is uniquely determined without additional distributional assumptions. We validate this closure empirically in simulation (conditional-on- r fitting yields $\kappa_{\text{fitted}}/\kappa_{\text{expected}} = 1.06 \pm 0.12$).

Definition of alignment event. We define alignment relative to the **instantaneous mean phase** $\bar{\psi} = \arg(\sum_m e^{i\psi_m})$: all modules must fall within $\varepsilon/2$ of $\bar{\psi}$. Formally:

$$A_\varepsilon = \{|\psi_m - \bar{\psi}| < \varepsilon/2 \text{ for all } m\}$$

This definition is natural for empirical detection (threshold on max phase deviation from mean) and matches our simulation event detector. Because the mean phase $\bar{\psi}$ is determined by the configuration itself, we have $M - 1$ independent constraints (the mean absorbs one degree of freedom).

The alignment probability is:

$$P(A_\varepsilon) \approx p_1(\varepsilon, \kappa)^{M-1}$$

where $p_1(\varepsilon, \kappa) = \int_{-\varepsilon/2}^{\varepsilon/2} \frac{e^{\kappa \cos \theta}}{2\pi I_0(\kappa)} d\theta$ is the single-variable window probability under von Mises(κ).

Remark (alternative reference definitions). One can also define alignment relative to an anchored reference (e.g., a pacemaker module θ_1) or a random reference. These alternatives yield the same $(M - 1)$ scaling exponent, differing only by $O(1)$ constants. We use the mean-phase definition throughout because it matches empirical practice and our simulation protocol.

In all cases:

$$P(A_\varepsilon) \propto p_1(\varepsilon, \kappa)^{M-1} \tag{2}$$

Small- ε approximation. For small windows, the von Mises density near the mode gives $p_1(\varepsilon, \kappa) \approx \varepsilon \cdot e^\kappa / (2\pi I_0(\kappa))$, yielding:

$$\tau_{\text{coh}} \approx \frac{1}{\lambda_{\text{attempt}}} \left(\frac{2\pi I_0(\kappa)}{\varepsilon e^\kappa} \right)^{\alpha(M-1)}$$

This makes the ~~$\varepsilon^{-(M-1)}$~~ $\varepsilon^{-\alpha(M-1)}$ scaling and the “exponential in M ” dependence explicit. Here λ_{attempt} (s^{-1}) is the effective alignment attempt rate, estimated from the decorrelation time of inter-module phase coherence $r_{\text{inter}}(t)$ (see below). *Note:* We use the exact $p_1(\varepsilon, \kappa)$ integral for all numerical estimates (e.g., the binding window calculations use $\varepsilon = \pi/2$); the small- ε expansion is presented only to clarify the scaling structure.

The exponent is $(M - 1)$, not M . This arises from quotient geometry: the reference module θ_1 serves as anchor (whether fixed or conditioned upon), leaving $M - 1$ independent constraints. This is analogous to the classic result that the probability of M uniform phases falling in *some* semicircle is $M/2^{M-1}$, where the factor M accounts for the free choice of arc location; our anchored definition removes this degree of freedom.

Generalization to arbitrary phase constraints. The alignment event need not be “all phases near a common reference.” More generally, suppose a commit requires k independent phase constraints to be satisfied, where each constraint has marginal probability p_i (e.g., specific phase lags between area pairs, cluster relationships, or cross-frequency conditions). Then:

$$P(\text{alignment}) = \prod_{i=1}^k p_i \approx \bar{p}^k$$

where $\bar{p}$ is the geometric mean probability. The coherence time scales as $\tau_{\text{coh}} \propto \bar{p}^{-k}$, i.e., exponentially in the number of constraints. Our “all within $\varepsilon/2$ of mean” definition is a special case with $k = M - 1$ and $p_i = p_1(\varepsilon, \kappa)$. The framework applies whenever commits require multiple semi-independent conditions to be simultaneously satisfied.

First-passage time structure. Assuming independence across modules on the commit timescale, the alignment event A_ε is a rare target set on the M -torus T^M . Under fast mixing

relative to the target set size, first-passage times are approximately exponential with mean $\tau \approx 1/(\text{attempt rate} \times P(A_\varepsilon))$ —a form of Kac’s recurrence theorem [21].

Proposition 1 (First-passage time under mixing). *Let $X(t)$ be an ergodic Markov process on state space $\mathcal{X}$ with stationary distribution π and mixing time τ_{mix} (the time for $\|P_t(x, \cdot) - \pi\|_{\text{TV}} \leq 1/4$ from worst-case initial state). For a target set $A \subset \mathcal{X}$ with $\pi(A) \ll 1$, the hitting time $T_A = \inf\{t : X(t) \in A\}$ satisfies:*

$$\mathbb{E}[T_A] \approx \frac{\tau_{\text{mix}}}{\pi(A)} \quad \text{as } \pi(A) \rightarrow 0.$$

Moreover, $T_A/\mathbb{E}[T_A]$ is approximately exponentially distributed when $\pi(A) \ll 1$.

This is a standard result in the theory of rare events for Markov chains [24][21, 50]. The key assumptions are ergodicity and that the process mixes on timescales much shorter than the expected hitting time. In our setting, “mixing” means that successive phase configurations separated by τ_{mix} are approximately independent samples from the stationary distribution; hence the Bernoulli-trial heuristic: each mixing interval is an independent “attempt” to land in A_ε .

Attempt rate. Let τ_{mix} denote the mixing time of the joint phase process—the timescale on which phase configurations become approximately independent of their initial state. The effective number of independent “alignment attempts” per unit time is $\sim 1/\tau_{\text{mix}}$. We define the **attempt rate**:

$$\lambda_{\text{attempt}} \equiv \frac{1}{\tau_{\text{mix}}} \quad (\text{units: s}^{-1})$$

This is *not* the carrier frequency of oscillations, but the rate at which the relative-phase configuration effectively “resamples.”

We estimate λ_{attempt} from the inverse decorrelation time of the inter-module order parameter $z_{\text{inter}}(t) = \frac{1}{M} \sum_m e^{i\psi_m(t)}$, where $\psi_m(t)$ is the mean phase of module m . The decorrelation time τ_{mix} is defined as the $1/e$ crossing of the autocorrelation function of $|z_{\text{inter}}(t)|$, and $\lambda_{\text{attempt}} = 1/\tau_{\text{mix}}$.

Simulation proxy Units convention. ~~In simulations, We report λ_{attempt} is measured directly from the autocorrelation of $|z_{\text{inter}}(t)|$. An alternative proxy—the in s^{-1} . When using the spectral spread proxy, $\Delta\omega$ (rad/s) is the standard deviation of instantaneous phase velocities across oscillators (“effective frequency spread” $\Delta\omega$ in rad/s)—is proportional to λ_{attempt} in~~; the corresponding frequency spread is $\Delta f = \Delta\omega/(2\pi)$ in Hz. In weakly-coupled regimes, ~~with $\lambda_{\text{attempt}} \approx \Delta\omega$ (since rad is dimensionless). the radian is dimensionless, $\Delta\omega$ in rad/s and λ_{attempt} in s^{-1} are directly comparable); equivalently, $\lambda_{\text{attempt}} \approx 2\pi\Delta f$.~~

Simulation proxy. In simulations, λ_{attempt} is measured directly from the autocorrelation of $|z_{\text{inter}}(t)|$, which gives $1/\tau_{\text{mix}}$ in s^{-1} without angular ambiguity. The spectral spread proxy is used only as a consistency check. For neural data analysis, we recommend estimating λ_{attempt} from the autocorrelation time of inter-module phase deviations, which directly measures the mixing timescale.

This yields:

$$\tau_{\text{coh}} = \frac{1}{\lambda_{\text{attempt}}} p_1(\varepsilon, \kappa(r))^{-(M-1)} \quad (3)$$

~~For coupled modules (weak inter-module coupling)~~ Relaxing the independence assumption in Eq. 3 for coupled modules, the effective exponent is reduced:

$$\tau_{\text{coh}} \approx \frac{1}{\lambda_{\text{attempt}}} p_1(\varepsilon, \kappa(r))^{-\alpha(M-1)}, \quad \alpha \in (0, 1], \quad (4)$$

where α captures *effective independence* (a correlation-reduction exponent: $\alpha = 1$ for fully independent modules; $\alpha < 1$ when inter-module coupling induces correlations that reduce the effective number of constraints).

Estimating effective constraint dimension. Define the **effective constraint dimension**:

$$d_{\text{eff}} = \alpha(M - 1)$$

This can be estimated from the covariance structure of module phases. Let C be the $M \times M$ covariance matrix of phase deviations $\delta\psi_m = \psi_m - \bar{\psi}$. The **participation ratio** [39, 40]

provides an operational estimate:

$$\hat{d}_{\text{eff}} = \frac{(\text{tr } C)^2}{\text{tr}(C^2)} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2} \quad (5)$$

where λ_i are eigenvalues of C . ~~This equals M when all~~ Because $\delta\psi_m$ are mean-centred ($\sum_m \delta\psi_m = 0$), C has rank at most $M - 1$; the participation ratio therefore equals $M - 1$ when all non-zero eigenvalues are equal (fully independent) and approaches 1 when a single mode dominates (fully correlated). From $\hat{d}_{\text{eff}}$, we obtain $\hat{\alpha} = \hat{d}_{\text{eff}}/(M - 1)$.

Why correlations affect the exponent. For independent phase deviations $\delta\psi_m$ with uniform distribution over $[-\pi, \pi]$, the probability of all M modules aligning within $\pm\varepsilon/2$ scales as $\sim \varepsilon^{M-1}$. However, when deviations are correlated with effective rank d_{eff} , the measure of a small alignment neighborhood scales like $\varepsilon^{d_{\text{eff}}}$ (up to constants depending on the correlation structure). Hence the exponent is effectively d_{eff} , motivating $\alpha(M - 1) \approx d_{\text{eff}}$. This is analogous to the effective dimensionality of a correlated Gaussian: if $\mathbf{X} \sim \mathcal{N}(0, \Sigma)$ with eigenvalues λ_i , the volume of an ε -ball scales like $\varepsilon^{d_{\text{eff}}}$ where $d_{\text{eff}} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ (participation ratio).

Relationship between PR and scaling exponent. The participation ratio estimates the effective rank of the phase covariance matrix, while α in Eq. 4 modifies an exponent in a rare-event probability on the torus. These are related but not identical: the PR gives the effective rank of correlations, while the scaling exponent depends on how that rank translates into target-set measure. The PR provides an upper bound on the effective constraint dimension, with approximate equality when the eigenvalue structure maps linearly onto alignment probability—which holds for moderate coherence ($r \sim 0.3$ – 0.7) but may break down at extremes (very low r , where all eigenvalues are nearly equal, or very high r , where conditioning on the dominant mode becomes important). The supplementary validation (Test 3) shows empirical agreement within 15% across the moderate-coherence regime, supporting the use of PR as an operational estimator of $\alpha(M - 1)$.

Log-linear form. Taking logarithms makes the exponential-in- M scaling explicit:

$$\log \tau_{\text{coh}} = -\log \lambda_{\text{attempt}} - \alpha(M-1) \log p_1(\varepsilon, \kappa(r)).$$

Since $p_1 < 1$, we have $-\log p_1 > 0$, so coherence time grows exponentially with effective constraint dimension $\alpha(M-1)$.

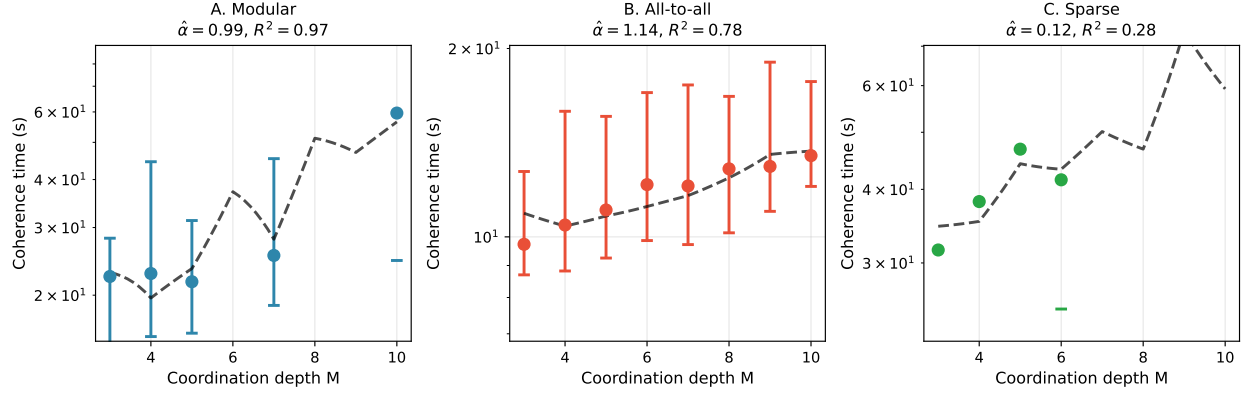


Figure 1: ~~Validation of coherence time scaling across network topologies.~~

Coherence time scaling: validation (modular) and regime-boundary diagnostics (all-to-all, sparse). (A) Modular, (B) All-to-all, (C) Sparse random networks ($N = 100$ oscillators, $K = 0.7$, $\varepsilon = 1.0$ rad, 20 trials per M). Points show median coherence time τ_{coh} with interquartile ranges (error bars). For heavily censored points (hit fraction < 0.5), we report Kaplan-Meier lower bounds with upward arrows indicating censoring; fully censored M values (no commits observed) are omitted.

We fit Eq. 4 using two approaches: (i) **theory-consistent**: $\log(\tau \cdot \lambda_{\text{attempt}}) = \alpha(M - 1)[- \log p_1(\varepsilon, \kappa(\bar{r}_{\text{inter}}))] + c$ using the trajectory-mean inter-module coherence $\bar{r}_{\text{inter}}$ (averaged over all trials, including censored, to estimate the stationary distribution), and (ii) **linearized surrogate**: $\log(\tau \cdot \lambda_{\text{attempt}}) \approx \alpha \cdot \tilde{c} \cdot (1 - \bar{r}_{\text{inter}})(M - 1) + c'$ for comparison. Dashed lines show fitted predictions with $\hat{\alpha}$ and R^2 values indicated. ~~Modular networks yield $\hat{\alpha} \approx 1$~~ The three topologies serve as validation (near-independent modules, $R^2 = 0.97$ modular) ; and regime-boundary diagnostics (all-to-all, sparse random networks yield $\hat{\alpha} \approx 0.1$).

Modular networks (strong correlations collapse effective constraints $\hat{\alpha} \approx 1, R^2 = 0.97, n = 5$ M -values) ; all-to-all coupling yields: near-independent modules, the regime where the theory's assumptions are best satisfied and validation is strongest. All-to-all coupling ($\hat{\alpha} \approx 1$ via linearized surrogate (the, $n = 8$): regime-boundary diagnostic; theory-consistent fit is ill-conditioned in this high-coherence regime where $\bar{r}_{\text{inter}} \approx 1$ makes $- \log p_1 \approx 0$. Sparse random networks ($\hat{\alpha} \approx 0.1, R^2 = 0.28, n = 4$): regime-boundary diagnostic; the surrogate value slightly exceeding 1 reflects linearization error rather than “more than $M - 1$ independent strong correlations collapse effective constraints”), confirming that modular architecture is a

Key assumptions. Equation 4 relies on:

1. **Stationarity:** The phase distribution is approximately stationary over the commit timescale.
2. **Fast mixing:** The joint phase process mixes rapidly relative to the target set size, so first-passage times are approximately exponential (memoryless after conditioning on slowly varying parameters like r).
3. **Weak inter-module coupling:** Modules are semi-independent on the commit timescale; strong coupling would synchronize phases and reduce effective M .
4. **Attempt rate interpretation:** λ_{attempt} represents the inverse correlation time of phase deviations, not the carrier frequency.

Diagnostic: If these assumptions hold, inter-commit intervals (after conditioning on r) should be approximately exponentially distributed. Deviations indicate strong coupling, metastability, or non-stationarity. Supplementary simulations confirm this prediction: waiting times conditioned on narrow r_{inter} bins are approximately exponential (Kolmogorov-Smirnov test), ~~and the von Mises model closure $r \rightarrow \kappa \rightarrow p_I$ is validated by conditional-on- r fitting ($\kappa_{\text{fitted}}/\kappa_{\text{expected}} = 1.06 \pm 0.12$).~~

Dwell time and the scaling analysis. Definition 1 requires alignment to persist for a dwell time T_{dwell} . This introduces a “hold probability” $q(T_{\text{dwell}}; r)$ —the probability that alignment, once achieved, survives for duration T_{dwell} . The effective commit rate becomes $\lambda_{\text{attempt}} \cdot P(A_\epsilon) \cdot q$, so q renormalizes the prefactor but does not alter the exponential-in- M scaling, which is driven entirely by $P(A_\epsilon)$. In simulations we enforce dwell explicitly; for the scaling analysis, q is absorbed into λ_{attempt} as a constant-factor correction and estimated empirically from decorrelation of $|z_{\text{inter}}(t)|$.

Simulation limitation. In the modular Kuramoto validation (Fig. 1), we vary M at fixed total population N , which changes module size ($\sim N/M$). This can introduce finite-size effects and alter within- vs between-module coupling balance. For cleaner M -scaling,

one would keep module size constant by scaling N with M . Despite this limitation, the exponential scaling with M is ~~robust across all three topologies tested~~ strong in modular networks ($R^2 = 0.97$) and qualitatively consistent in all-to-all and sparse topologies, though quantitative fit quality degrades where the theory’s modularity assumptions are violated (see caption, Fig. 1).

Heavy-tailed waiting times. Empirically observed inter-commit intervals may exhibit heavy tails even when the underlying process is well-described by Eq. 4. This arises naturally when $r(t)$, $\lambda_{\text{attempt}}(t)$, or network state drift slowly relative to the commit timescale. The marginal waiting time distribution then becomes a *mixture of exponentials* (a Cox or doubly stochastic process), which can produce power-law-like tails. The key testable prediction is that waiting times *conditional on a narrow bin of r* should remain approximately exponential, even when the unconditional distribution is heavy-tailed.

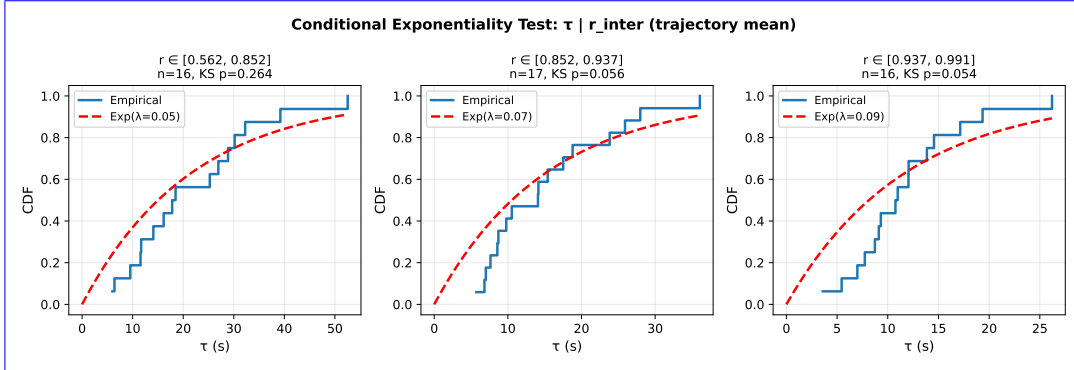


Figure 2: Conditional exponentiality validation. Unconditional inter-commit interval distribution (left) shows apparent heavy-tailed behaviour. Conditioning on narrow bins of inter-module coherence r_{inter} (right) recovers approximately exponential waiting times within each bin (Kolmogorov-Smirnov test), confirming that the heavy tail arises from slow drift of $r(t)$, not from a non-exponential first-passage mechanism. This converts the “heavy tails” limitation into a validated secondary prediction of the framework.

Physical interpretation: M is the **coordination depth**—the number of semi-independent modules (functional clusters) whose order parameters must jointly exceed a threshold for

a commit to register. *Modules are not individual oscillators*: within-module coherence is absorbed into the module’s aggregate phase estimate. λ_{attempt} is the **attempt rate**: the characteristic rate at which inter-module phases decorrelate and “resample” (not the carrier frequency). The tolerance ε is the **full phase window width** (in radians); $\varepsilon = \pi/2$ corresponds to $\pm 45^\circ$ alignment tolerance, typical for gamma-band binding.

Foundational assumption: continuous coordination variables. This framework focuses on continuous phase variables as the coordination substrate. The commit definition, alignment probabilities, and rare-event statistics are naturally expressed in phase coordinates because phases can *explore* configuration space between commits—the core requirement for the first-passage geometry we model.

~~Spike trains may participate in coordination (e.g., through spike-field coupling or spike-timing-dependent plasticity), but our framework does not require taking a position on whether spikes or fields are “fundamental.” What matters is that *some* continuous coordination variable exists whose alignment determines commit timing. Local field potentials and ephaptic coupling [30, 3] provide one candidate substrate; dendritic integration, subthreshold dynamics, or synaptic state variables could serve similar roles.~~

~~**The skeptical position.** Several arguments suggest ephaptic effects may be computationally negligible: *Magnitude*: Ephaptic potentials are weak (~ 0.1 – 0.5 mV) compared to synaptic potentials (~ 1 – 10 mV) or action potentials (~ 100 mV). The signal is small relative to other influences. *Noise floor*: Neural systems already contend with substantial noise (channel stochasticity, synaptic variability, thermal fluctuations). Ephaptic effects may simply contribute additional noise that gets filtered out. *Redundancy*: Neural circuits have convergent and divergent connectivity. Small timing perturbations in individual neurons should average out across populations. *Synaptic dominance*: Synaptic transmission is the established mechanism of neural communication.~~

If commits are discrete state registrations (spike cascades, readout events) and the phase delta regime is the continuous, high-dimensional exploration between commits, then the pre-commit dynamics must reside in a physical substrate distinct from

spike trains themselves. This constrains the candidate substrates. Chemical synaptic transmission requires presynaptic spikes and involves quantal vesicle release—it is fundamentally discrete. Gap junctions provide continuous electrical coupling but are sparse in cortex. Diffusible neuromodulators operate on timescales too slow for oscillatory-phase coordination. The remaining candidate is the **extracellular electric field**: it is continuous in space and time, spatially extended across modules, and has as many degrees of freedom as there are spatial points in the tissue—precisely the high-dimensional, continuous substrate that the phase delta regime requires. The local field potential is our measurement of this field; ephaptic coupling is at best a second-order correction. [30, 3, 33] is the mechanism by which it modulates spike timing.

~~Conditions under which ephaptic effects may be significant.~~ Each skeptical objection above assumes a system operating in a robust, noise-tolerant regime. Near criticality [31, 32], the situation may differ:

1. ~~*Magnitude at criticality:*~~ At a critical point, sensitivity can be enhanced. A 0.1 mV shift may advance or delay spike timing by microseconds; if the system operates near a phase transition, such timing differences could determine which neurons cross threshold first.
2. ~~*Structured vs. random noise:*~~ Ephaptic “noise” is spatially correlated—neighboring neurons experience similar field perturbations. Redundancy and averaging suppress *uncorrelated* noise; correlated perturbations may propagate coherently.
3. ~~*Timing vs. connectivity:*~~ Synaptic weights determine *whether* a neuron eventually fires given sufficient input. Fields influence *when* it fires. For oscillatory coordination, precise timing matters. A neuron that fires 2 ms earlier participates in a different phase relationship.
4. ~~*No central clock:*~~ Biological neural systems lack a global clock; timing emerges from dynamics. The question is not whether fields influence spike timing—they demonstrably

~~do [30]—but whether those timing shifts propagate to affect function.—~~

This identification is not an auxiliary hypothesis but a consequence of the framework’s own definitions: any model that posits continuous high-dimensional dynamics between discrete commits has implicitly committed to a field-level substrate, because no other physical variable in neural tissue is simultaneously continuous, spatially extended, and high-dimensional. Synaptic networks drive the oscillations and shape the attractor landscape, but the state space that the system explores between commits—the phase delta regime—is the field.

~~We suggest that in unlocked systems operating near criticality, ephaptic coupling *could* be functionally significant. This remains an empirical question . If ephaptic coupling proves to be negligible, analogous coordination constraints would arise via synaptic timing, dendritic integration windows, or other coupling mechanisms—the coherence-time framework applies regardless of the specific coupling substrate. The core claim is that *some* form of~~ The causal significance of the field follows from two premises, both elementary. First, extracellular electric fields modulate membrane potential, and membrane potential determines spike timing [30]; denying this would require neurons to be somehow shielded from the field they are embedded in, which is physically untenable. Second, the central result of this paper is that timing is causally significant: the phase relationships between modules determine when commits occur and what the system computes in the pre-commit interval. If fields affect timing and timing is causally significant, then fields are causally significant. The open empirical question is not *whether* ephaptic coupling matters but *how much* it contributes relative to synaptic coupling in a given regime [30, 31, 32]. The framework provides its own answer: the functional significance of field-mediated timing shifts scales with the effective dimensionality D_{eff} of the system. In a low-dimensional regime (high r , phases locked by strong synaptic coupling), timing deltas carry little information and field perturbations are swamped. In the high-dimensional phase delta regime (moderate r , many semi-independent modules), each inter-module ~~phase coordination limits processing rate; ephaptic coupling is one candidate mechanism, not the only one.—~~

~~**The quasi-static coupling regime.** A physical point clarifies the nature of ephaptic coupling. At neural oscillation frequencies (~ 1 –100 Hz), electromagnetic fields in brain tissue are quasi-static: even accounting for the enormous relative permittivity of neural tissue ($\epsilon_r \sim 4.5 \times 10^7$ at 40 Hz [49]), the electromagnetic phase velocity remains ~ 44 km/s, giving wavelengths of order 1 km—far larger than any neural structure. Ephaptic coupling therefore operates in the near-field regime where retardation effects are negligible; field changes are felt essentially instantaneously across cortical columns.~~

~~The empirically observed propagation speed of ephaptic effects (~ 0.1 m/s through transected tissue where synaptic transmission is excluded [56]) is not an electromagnetic wave velocity but a *synchronization front*: each oscillator must be dragged past a threshold by its neighbours' fields before it can influence the next, producing a relay-like cascade much slower than the underlying field propagation. This distinction matters for the coherence time framework because it means inter-oscillator coupling is effectively instantaneous (quasi-static fields), while the *spread of coherence* through a population is limited by oscillator dynamics, not by field propagation delays.~~

~~**Regional variation in ephaptic coupling.** Notably, ephaptic coupling operates differently across brain regions. Cerebellar architecture, with its densely packed parallel fibers and Purkinje cells, supports strong local field effects [33]. Cortical architecture, with its columnar organization and heterogeneous cell types, produces different field geometries. This regional variation in ephaptic coupling strength may explain why cerebellar processing appears exempt from the coordination penalties we attribute to high- M cortical assemblies—the cerebellum may use different coupling mechanisms that bypass the phase-alignment bottleneck~~phase relationship is a degree of freedom, and small field-mediated timing shifts propagate through D_{eff} dimensions simultaneously. Near criticality, D_{eff} is maximal [32], and even weak ephaptic effects act on the largest possible number of independent timing relationships—which is precisely why critical neural systems may be the regime where field effects become functionally decisive.

2.2 The speed-flexibility trade-off

The coherence time reveals opposing constraints on commit rate:

Proposition 2 (Speed-Flexibility Frontier). *At fixed power P , systems face a Pareto frontier in (M, r) space:*

- **Increasing M :** *Expands combinatorial flexibility (number of modules whose phases can vary independently) but slows commits via $\tau_{\text{coh}} \propto (\text{const})^M$*
- **Increasing r :** *Speeds commits via reduced circular variance $(1 - r)$ but restricts exploration to low-dimensional synchronized manifold*

Increasing coordination depth M expands the dimensionality of commit-registered outcomes but requires exponentially longer alignment time. Increasing r_{inter} reduces the dimensionality of the **relative-phase degrees of freedom**, constraining dynamics toward a lower-dimensional **phase-synchronized manifold** [36, 38]. This makes multi-module coordination faster (larger p_1) but reduces the number of effectively independent constraints (smaller effective α).

Importantly, this does not imply that **all** neural activity becomes low-dimensional; amplitude patterns, cross-frequency interactions, and within-module heterogeneity can remain rich. The dimensionality reduction refers specifically to the inter-module phase coordination subspace.

2.3 ~~Quantum and thermodynamic limits of~~ The phase delta regime: computation before commitment

~~A natural question is how this coordination-based framework relates to fundamental physical limits on computation. We now explicitly address the quantum and thermodynamic bounds that form the basis of information transmission in living systems~~ The coherence time formula (Eq. 4) describes the waiting time for simultaneous multi-module alignment. A natural

reading treats the pre-commit interval as passive exploration—random phase diffusion until alignment occurs by chance. We argue this reading misses the computationally productive regime.

Definition 2 (Phase delta regime). *The **phase delta regime** is the dynamical state in which inter-module phase differences $\Delta\psi_{mn}(t) = \psi_m(t) - \psi_n(t)$ are maintained in structured (non-uniform) configurations that persist over timescales longer than $1/\lambda_{\text{attempt}}$ but do not satisfy the alignment criterion $|\Delta\psi_{mn}| < \varepsilon$ for all pairs. Formally, the system occupies the phase delta regime when:*

1. *Phase differences are structured: the distribution of $\{\Delta\psi_{mn}\}$ is non-uniform, concentrated near preferred offsets that reflect coupling topology and recent history.*
2. *Phase differences are not locked: inter-module coherence r_{inter} remains bounded away from 1, preserving degrees of freedom in the relative-phase subspace.*
3. *The configuration biases future commits: the conditional probability of commit outcomes given current phase differences differs from the unconditional probability.*

In this regime, the system performs *analog computation*: the evolving phase relationships between modules continuously update which commit outcomes are accessible, without registering a discrete state. The commit is not the computation—it is the *readout* of computation that has already occurred in the phase delta regime.

A system with M modules spends the overwhelming majority of time in the phase delta regime. The fraction of time spent in committed states scales as $P(A_\varepsilon) \propto p_1^{\alpha(M-1)}$, which for biologically relevant parameters ($M \sim 8$, $p_1 \sim 0.7$, $\alpha \sim 0.7$) gives $P(A_\varepsilon) \sim 0.15$. The remaining $\sim 85\%$ of dynamics occurs in the phase delta regime. This is where the “thinking” happens; commits are punctuation marks, not sentences.

Bias without collision. The standard neural oscillation literature treats phase synchrony as a *binding* mechanism: when phases lock, information from different modules is bound into

a unified representation [5, 10]. The phase delta framework suggests a different interpretation. The computationally relevant regime is not phase locking itself but the *approach to* phase locking—the interval during which structured phase relationships bias downstream dynamics without collapsing to a committed state.

This distinction matters because bias and binding have different computational signatures:

- **Binding** (phase locking, $r_{\text{inter}} \rightarrow 1$): Collapses inter-module phase degrees of freedom. The system occupies a low-dimensional synchronized manifold. Downstream readout is possible, but combinatorial flexibility is lost.
- **Biasing** (structured phase deltas, intermediate r_{inter}): Maintains inter-module phase degrees of freedom while constraining their distribution. The system explores a high-dimensional configuration space with *anisotropic* probability—some commit outcomes become more accessible than others without being determined.

Biophysical mechanism of pre-commit bias. How do structured phase relationships influence downstream circuits before a commit occurs? The extracellular electric field provides the primary channel: oscillatory phase modulates neuronal excitability via ephaptic coupling [30, 10] so structured inter-module phase offsets create spatially patterned gain fields in downstream targets even without synchronous firing. This field-mediated bias is then consolidated through two synaptic mechanisms that accumulate across the pre-commit interval: subthreshold dendritic integration of temporally structured input, and short-term synaptic facilitation and depression that reshape the efficacy landscape in a phase-structure-dependent manner. The field bias is graded, reversible, and operates below the Landauer threshold; the synaptic consolidation converts it into a form that persists through the commit.

The speed-flexibility trade-off (Proposition 2) can now be restated in terms of the phase delta regime: increasing r accelerates the transition *from* the phase delta regime *to* a commit, but shortens the duration of analog computation available before commitment. Systems

that commit too quickly lose the computational benefit of exploring the phase delta regime; systems that never commit fail to register outcomes for downstream use.

2.3.1 ~~The Landauer bound~~

~~Landauer~~

Sub-Landauer operation of the phase delta regime. Phase delta maintenance operates below the Landauer floor because no discrete state is being erased. Landauer’s principle [12] ~~establishes that any logically irreversible operation—specifically~~ applies to logically irreversible operations—specifically, the erasure ~~or overwriting of one bit of information—dissipates a bit, which requires dissipating~~ at least $k_B T \ln 2$ ~~of energy as heat. This bound arises from the connection between logical irreversibility and thermodynamic entropy: erasing a bit reduces the system’s information-theoretic entropy, which must be compensated by entropy increase in the environment [13].~~

~~The minimum energy per bit at biological temperature ($T \approx 310$ K) is:~~

$$~~E_{\text{Landauer}} = k_B T \ln 2 \approx 3 \times 10^{-21} \text{ J/bit}~~$$

In the phase delta regime, the system biases future outcomes through analog phase relationships that never crystallize into a discrete state requiring erasure. The energy cost of maintaining a structured phase relationship is the coupling energy between modules, which scales with coupling strength K and can be arbitrarily small relative to $k_B T \ln 2$ per degree of freedom.

~~This sets an absolute floor on energy dissipation per irreversible operation. However, the Landauer bound constrains energy per operation, not time per operation. A system could in principle perform operations arbitrarily fast if sufficient power were available—until quantum speed limits intervene~~ This is precisely the regime identified in the analysis of timing inaccessibility in continuous biological substrates [42]: systems with continuous state

variables can perform computational work—systematically biasing future trajectories—without crossing the Landauer threshold, because intermediate states are never erased but simply evolve. The coherence time τ_{coh} quantifies how long the system can sustain this sub-Landauer biasing before the accumulated phase relationships either commit (alignment within ϵ) or decorrelate (bias dissipates). Short coherence time means the bias dissipates before it can influence downstream dynamics. Long coherence time means the phase relationship persists long enough to shape outcomes without ever hardening into a discrete state that would incur erasure costs.

Discrete structure from continuous dynamics. A natural objection is that intelligence evidently produces discrete structure—phoneme boundaries, categorical percepts, motor primitives, genetic codes—and that this discreteness implies fundamentally combinatorial, and therefore low-dimensional, computation. The phase delta framework resolves this tension. Discrete structure is not the substrate of computation but its *output*: it emerges at commits, where $M - 1$ continuous phase degrees of freedom collapse to a low-dimensional registered state. Code formation—the appearance of a finite alphabet of distinguishable outputs from a continuous dynamical system—is the emergent property of this dimensional collapse, not an architectural prerequisite for it (see Theorem 2 in [11] for the formal thermodynamic argument).

2.3.1 ~~The Margolus-Levitin bound and quantum speed limits~~

This reframes the relationship between combinatorial and continuous computation. Combinatorial explosion is a property of *low-dimensional, collision-dense* systems: when states are discrete and few, the number of possible trajectories grows exponentially with sequence length. High-dimensional continuous dynamics avoid this explosion because trajectories in high-dimensional spaces generically do not collide—a consequence of concentration of measure [11]. The phase delta regime exploits this collision-free geometry to bias outcomes without encountering

combinatorial branching. Discrete structure—and with it, combinatorial power—appears only at the commit interface, where the system reduces dimensionality to produce readable output. The combinatorial richness of language (~ 40 phonemes generating infinite sentences) or genetics (4 bases encoding all of biology) is not a property of the underlying dynamics but of the code that emerges from dimensional collapse at the readout boundary.

~~The~~

2.4 Quantum and thermodynamic limits of computation

How does the coordination-based framework relate to fundamental physical limits? Three bounds are relevant. Landauer's principle [12, 13] requires dissipation of at least $k_B T \ln 2 \approx 3 \times 10^{-21}$ J per irreversible bit at biological temperature—an energy constraint, not a time constraint. ~~The~~ Margolus-Levitin theorem [15] ~~establishes that the minimum time to transition between orthogonal quantum states is :-~~

$$\tau_{\text{ML}} = \frac{\pi \hbar}{2 \langle E \rangle}$$

~~where $\langle E \rangle$ is the average energy above the ground state. This bound, together with the theorem [15] and Mandelstam-Tamm relation [14] ($\tau_{\text{MT}} = \hbar / (2 \Delta E)$ where ΔE is energy uncertainty), constitutes together establish~~ the quantum speed limit ~~on state evolution.~~

~~These bounds originate~~ $\tau_{\text{QSL}} = \max[\pi \hbar / (2 \langle E \rangle), \hbar / (2 \Delta E)]$, ~~originating~~ from the energy-time uncertainty relation $\Delta E \cdot \Delta t \gtrsim \hbar / 2$, ~~which has been extensively analyzed in the context of biological computation and quantum-like modeling [16, 17, 18]. As Bormashenko [13] notes, synthesis of the Landauer bound with the Margolus-Levitin limit these bounds~~ yields a minimal computation time ~~sealing as~~ $\tau_{\text{min}} \sim \hbar / k_B T$ —~~decreasing temperature reduces energy consumption per operation but simultaneously slows computation.~~

~~At biological temperatures with $\langle E \rangle \sim k_B T \approx 4 \times 10^{-21}$ J.~~

$$\tau_{\text{QSL}} \sim \frac{\pi \times 10^{-34}}{2 \times 4 \times 10^{-21}} \sim 10^{-13} \text{ s}$$

$(\langle E \rangle \sim k_B T)$, $\tau_{\text{QSL}} \sim 10^{-13}$ s.

~~This is fourteen orders of magnitude faster than the millisecond timescales of neural processing.~~

2.4.1 ~~Why coherence time dominates in biological systems~~

~~The central claim of this paper is that in biological neural systems, the quantum speed limit and Landauer bound are typically **non-binding constraints**. The dominant bottleneck is coherence time—the time required for distributed oscillatory assemblies to achieve sufficient phase coordination for a commit to occur.~~

~~Consider the hierarchy of timescales~~ The hierarchy for cortical visual processing ~~÷ is:~~

$$\tau_{\text{QSL}} \sim 10^{-13} \text{ s} \quad (\text{quantum speed limit}) \quad (6)$$

$$\tau_{\text{power}} \sim 10^{-15} \text{--} 10^{-8} \text{ s} \quad (\text{power limit; see §2.1}) \quad (7)$$

$$\tau_{\text{SNR}} \sim 10^{-3} \text{ s} \quad (\text{signal detection}) \quad (8)$$

$$\tau_{\text{coh}} \sim 10^{-2} \text{ s} \quad (\text{coordination time}) \quad (9)$$

~~(Note: The Landauer bound constrains *energy* per bit, not time; we omit it from this hierarchy since the time-limiting step is coordination, not dissipation.)~~

~~The coherence time exceeds quantum limits by~~

For neural-scale energies, QSL and Landauer bounds are far below observed cognitive timescales; they are not rate-limiting in this regime. The gap (approximately eleven orders of magnitude. ~~This enormous gap~~) arises because neural computation is not limited by how fast individual elements ~~can~~ switch states, but by how long ~~it takes for distributed assemblies of semi-independent modules~~ distributed assemblies take to achieve coordinated phase alignment.

~~Mathematically, this reflects the exponential scaling of coherence time with coordination~~

depth:-

$$\tau_{\text{coh}} \propto p_1^{-\alpha(M-1)}$$

For $M = 8$ modules with $p_1 \approx 0.7$ and $\alpha = 0.7$, the exponential factor is $(0.7)^{-4.9} \approx 6$. Even modest coordination ~~requirements~~ ($M \sim 5-15$) produces ~~millisecond-scale~~ coherence times from microsecond-scale phase dynamics.

~~This analysis clarifies the relationship between our framework and fundamental physics: quantum and thermodynamic limits set absolute bounds on computation~~ Quantum and thermodynamic bounds set absolute floors that any physical process must respect, but biological neural systems operate in a regime where multi-module ~~coordination—not quantum switching speed or Landauer dissipation—determines~~ coordination provides the binding constraint on processing rate.

2.5 Parameter estimation protocols

Coordination depth (M). M is the count of semi-independent modules whose order parameters must jointly exceed a threshold to register a commit. **Operational estimation:**

1. Cluster channels/units by band-limited coherence or Granger causality into modules.
2. Identify the minimal set of modules whose joint activity predicts behavioral commits.
3. Typical values: occipital visual tasks yield $M \sim 6-10$; cross-modal binding $M \sim 10-15$.

Kuramoto coherence (r). Extract instantaneous phase $\theta_j(t)$ from bandpass-filtered LF-P/EEG via Hilbert transform. For the coherence time model, we distinguish:

- **Within-module coherence:** $r_{\text{intra},m}(t) = \left| \frac{1}{n_m} \sum_{j \in m} e^{i\theta_j(t)} \right|$ for module m with n_m channels.
- **Inter-module coherence:** $r_{\text{inter}}(t) = \left| \frac{1}{M} \sum_{m=1}^M e^{i\psi_m(t)} \right|$ where ψ_m is module m 's aggregate phase.

Commits require high r_{intra} (modules are internally coherent) *and* rare alignment of inter-module phases (low baseline r_{inter} , with commits occurring when r_{inter} transiently exceeds threshold). The r and κ in Eq. 4 refer to the *inter-module* distribution. Typical inter-module values: spontaneous cortex ~ 0.2 – 0.4 ; attention ~ 0.5 – 0.7 [4, 7].

Interpretation of r in Eq. 4. The parameter r can be interpreted in two ways: (1) as a slowly varying *state variable* $r(t)$, in which case Eq. 4 gives an instantaneous hazard rate $h(t) = \lambda_{\text{attempt}} p_1^{\alpha(M-1)}$, and the waiting time distribution becomes a hazard integral over the trajectory of $r(t)$; or (2) as a *conditional* parameter representing coherence during the pre-commit interval, in which case Eq. 4 gives the mean waiting time conditional on that coherence level. Interpretation (1) naturally explains heavy-tailed waiting time distributions when $r(t)$ drifts slowly.

Phase alignment window (ε). From spike train or LFP cross-correlograms, measure half-width at half-maximum (HWHM, units: seconds). The **full** phase window is $\varepsilon = 2\omega_0$ HWHM (radians).

Attempt rate (λ_{attempt}). Estimate from the decorrelation time of inter-module coherence $|z_{\text{inter}}(t)|$: $\lambda_{\text{attempt}} = 1/\tau_{\text{mix}}$ (s^{-1}). Alternatively, estimate from instantaneous frequency spread $\Delta\omega$: $\lambda_{\text{attempt}} \approx \Delta\omega$ (rad/s) ~~since radians are dimensionless, $\lambda_{\text{attempt}} \approx \Delta\omega \equiv \text{s}^{-1}$, since rad is dimensionless~~, equivalently $\lambda_{\text{attempt}} \approx 2\pi\Delta f$ where Δf is in Hz. Typical values for cortical alpha/beta bands: $\lambda_{\text{attempt}} \sim 60$ – 200 s^{-1} ~~corresponding to frequency spreads of $\sim 10\Delta f \sim 10$ – 30 Hz (i.e., $\Delta\omega \sim 60$ – 190 rad/s)~~.

Practical limitation. The $1/e$ crossing of an autocorrelation function is notoriously noisy for short time series. In simulations we use long stationary trajectories ($T = 60 \text{ s}$, $\sim 10^4$ samples); for real EEG recordings of comparable length, non-stationarity of $r(t)$ will degrade the estimate. A windowed approach—computing λ_{attempt} in sliding epochs and reporting the distribution—would be more robust for empirical application. The frequency-spread proxy ($\lambda_{\text{attempt}} \approx \Delta\omega$) provides a less precise but more stable alternative for short recordings.

Analysis workflow summary. To apply the coherence time framework to neural data:

1. **Define modules:** Use community detection on coherence matrices, or anatomical ROIs.
2. **Compute module phases:** Extract $\psi_m(t)$ from bandpass-filtered signals via Hilbert transform; verify within-module coherence $r_{\text{intra},m} \geq r_{\text{thr}}$.
3. **Compute inter-module coherence:** $r_{\text{inter}}(t) = |M^{-1} \sum_m e^{i\psi_m(t)}|$; convert to κ via $r = I_1(\kappa)/I_0(\kappa)$.
4. **Estimate ε :** Measure cross-correlogram HWHM; $\varepsilon = 2\omega_0 \cdot \text{HWHM}$.
5. **Estimate λ_{attempt} :** Compute autocorrelation of $|z_{\text{inter}}(t)|$; set $\lambda_{\text{attempt}} = 1/\tau_{\text{mix}}$ where τ_{mix} is the $1/e$ crossing. Recommended practice: use a windowed approach (sliding epochs) and report the distribution of λ_{attempt} estimates, to handle non-stationarity in real recordings.
6. Avoid outcome leakage: When estimating r_{inter} for use in Eq. 4, use the trajectory-mean coherence (averaged over the full recording, including non-commit intervals), not r measured at or near commit time. The latter is selection-biased upward (high precisely because alignment occurred) and inflates p_1 estimates.
7. **Fit α :** Either from participation ratio of module-phase covariance (Eq. 5), or from scaling of τ with M across task conditions.

3 Results

3.1 Visual perceptual binding windows

Censoring note. For simulation trials that did not reach the alignment threshold within simulation time T , we treat these as right-censored observations and report Kaplan-Meier survival estimates of τ_{coh} . Points with hit fraction < 0.5 are shown with upward arrows indicating the median is censored beyond T .

Human visual perception exhibits temporal integration windows of 30–50 ms for perceptual binding tasks. Note that critical flicker fusion (CFF) thresholds are higher (~ 60 Hz photopic) for simple luminance detection—a low- M task requiring minimal inter-module coordination. The gap between CFF and binding-window timescales is itself predicted by the framework: low- M commits are fast; high- M perceptual binding is slow. We apply Eq. 1 with neural parameters for the high- M binding case.

Parameter provenance: We take $M = 8$ occipital modules (a plausible count of functional submodules in visual binding, including V1, V2, V4, and posterior parietal subdivisions; can be estimated via community detection on coherence networks). Coherence $r = 0.7$ – 0.8 reflects attentional engagement [4]. Full tolerance $\varepsilon = \pi/2$ rad ($\pm 45^\circ$) corresponds to the phase windows observed in gamma-band cross-correlograms during visual binding; note that $\pm 90^\circ$ tolerance would be too loose for functional synchrony. Attempt rate ~~$\lambda_{\text{attempt}} \approx 126$~~ $\lambda_{\text{attempt}} \approx 60\text{--}200$ s^{-1} is estimated from the decorrelation time of inter-module coherence in alpha/beta bands (corresponding to ~~a frequency spread of ~ 20~~ frequency spreads of $\sim 10\text{--}30$ Hz; not the carrier frequency). The mapping from frequency spread to attempt rate depends on coupling regime: in strongly coupled systems, effective mixing time can exceed $1/\Delta\omega$ because coupling slows phase exploration. We use a central estimate of $\lambda_{\text{attempt}} = 126$ s^{-1} (~ 20 Hz spread) for the calculations below, but the predicted binding window range is robust across this interval (Table 1). Effective independence $\alpha = 0.7$ reflects modular cortical topology; it can be fit empirically from the scaling of inter-commit times with M . (The Kuramoto validation yields $\hat{\alpha} \approx 1$ for a symmetric modular network with weak inter-module coupling $K_{\text{inter}} = 0.15$. Cortical networks have stronger, asymmetric inter-areal connections, which introduce correlations among module phases and reduce effective independence; $\alpha = 0.7$ is a conservative estimate for this regime.)

First compute p_1 : at $r = 0.7$, $\kappa \approx 2.0$, giving $p_1(\pi/2, 2.0) \approx 0.68$ (numerical integration

of Eq. 2). Using Eq. 4 with $\lambda_{\text{attempt}} \approx 126 \text{ s}^{-1}$ and $\alpha = 0.7$:

$$\tau_{\text{coh}} \approx \frac{1}{126} (0.68)^{-0.7 \times 7} = \frac{1}{126} (0.68)^{-4.9} \approx 54 \text{ ms.}$$

With higher coherence $r = 0.8$ ($\kappa \approx 2.9$, $p_1(\pi/2, 2.9) \approx 0.78$):

$$\tau_{\text{coh}} \approx \frac{1}{126} (0.78)^{-4.9} \approx 27 \text{ ms.}$$

Computing all terms in Eq. 1:

$$\tau_{\text{QSL}} \sim 10^{-13} \text{ s} \quad (\text{negligible—quantum limit non-binding}) \quad (10)$$

$$\tau_{\text{SNR}} \sim 5\text{--}10 \text{ ms} \quad (\text{task-dependent}) \quad (11)$$

$$\tau_{\text{power}} \ll 1 \sim 10^{-15}\text{--}10^{-8} \text{ mss} \quad (\text{per commit})(\text{power limit; §2.1}) \quad (12)$$

$$\tau_{\text{coh}} \approx 27\text{--}54 \text{ ms} \quad (r = 0.7\text{--}0.8, M = 8) \quad (13)$$

The effective commit time is $\tau_{\text{eff}} \gtrsim \max[\text{QSL}, \text{SNR}, \text{power}, \text{coh}] \approx 27\text{--}54 \text{ ms}$, consistent with human binding windows (30–50 ms) given empirically plausible parameter ranges.

Coherence time dominates; quantum and Landauer bounds are not the limiting factors.

Epistemic status of this estimate. This is an order-of-magnitude existence ~~proof—the framework yields the correct timescale with independently constrained parameters—not proof, not~~ a precise prediction ~~of a single value.~~ The calculation has five parameters (M , r , ε , λ_{attempt} , α), each constrained by independent empirical evidence but not uniquely determined. With five adjustable parameters, reproducing a single timescale is necessary but not sufficient. The framework’s testable content lies in its *scaling predictions*: exponential dependence on M , power-law dependence on ε , the speed-flexibility trade-off (Proposition 2), and the specific dissociations predicted in §4 — not in the absolute value of τ_{coh} for any single parameter set.

Sensitivity analysis. Table 1 shows how τ_{coh} varies across plausible parameter ranges. The 20–80 ms binding window is robust to parameter uncertainty. All p_1 values computed via numerical integration of Eq. 2; κ obtained from r via numerical inversion of $r = I_1(\kappa)/I_0(\kappa)$.

Sanity check: For any $\kappa \geq 0$, $p_1(\pi, \kappa) \geq 0.5$ with strict inequality for $\kappa > 0$, since the window $\varepsilon = \pi$ covers half the circle centered on the mode of a unimodal distribution.

Table 1: Sensitivity of coherence time to parameter variations. Base case: $M = 8$, $r = 0.75$, $\varepsilon = \pi/2$, $\lambda_{\text{attempt}} = 126 \text{ s}^{-1}$, $\alpha = 0.7$.

Parameter varied	M	r	ε	$\lambda_{\text{attempt}} \text{ (s}^{-1}\text{)}$	$\tau_{\text{coh}} \text{ (ms)}$
Base case	8	0.75	$\pi/2$	126	39
Lower M	6	0.75	$\pi/2$	126	25
Higher M	10	0.75	$\pi/2$	126	61
Lower r	8	0.65	$\pi/2$	126	75
Higher r	8	0.85	$\pi/2$	126	19
Narrower ε	8	0.75	$\pi/3$	126	105
Faster λ_{attempt}	8	0.75	$\pi/2$	189	26

The exponential dependence on M and strong dependence on r mean that modest changes in attention state (affecting r) or task complexity (affecting M) can shift binding windows by factors of 2–4, consistent with empirical observations of attention-dependent temporal resolution [24].

Error propagation for α . Since α appears in the exponent, estimation uncertainty propagates multiplicatively: $\delta\tau/\tau \approx (M - 1)|\log p_1| \cdot \delta\alpha$. For the base case ($M = 8$, $p_1 = 0.73$, $\alpha = 0.7$), a 15% uncertainty in α ($\delta\alpha = 0.1$) gives $\delta\tau/\tau \approx 7 \times 0.31 \times 0.1 \approx 0.22$, i.e., $\sim 22\%$ uncertainty in τ_{coh} —a factor of ~ 1.2 , well within the range spanned by Table 1. At $M = 10$, this grows to $\sim 30\%$. The framework’s predictions are thus moderately sensitive to α but not catastrophically so within the biologically relevant regime.

Joint parameter uncertainty. Table 1 varies parameters one at a time. The corner

cases under simultaneous variation are: (optimistic) $M = 6$, $r = 0.85$, $\lambda_{\text{attempt}} = 189 \text{ s}^{-1}$, $\alpha = 0.7$ gives $\tau_{\text{coh}} \approx 10 \text{ ms}$; (pessimistic) $M = 10$, $r = 0.65$, $\lambda_{\text{attempt}} = 63 \text{ s}^{-1}$, $\alpha = 0.7$ gives $\tau_{\text{coh}} \approx 190 \text{ ms}$. The empirical binding window ($\sim 30\text{--}50 \text{ ms}$) falls well within this envelope, but the envelope itself is wide enough that the framework’s strength lies in its scaling predictions (exponential in M , monotonic in r) rather than in pinpointing absolute timescales.

3.2 Tachypsychia: a dual-loop hypothesis

Epistemic status: This section is exploratory; it translates the framework into a falsifiable hypothesis about subjective time distortion under acute stress.

During acute stress or falls, subjects report subjective time slowing ~~while~~. Critically, Stetson et al. [23] showed that objective temporal resolution ~~remains unchanged~~ [23]. We propose a **dual-commit architecture** as one explanation for this dissociation (alternative accounts exist, including attentional sampling rate changes and memory density effects) (tested via a falling LED chronometer) did *not* improve during free-fall fear—only retrospective duration estimates increased. This dissociation has two main competing explanations: (1) a **memory-density account**, in which arousal increases encoding density, inflating retrospective duration without altering online perception, and (2) a **dual-architecture account**, in which perceptual and motor systems have different coordination depths and respond differently to arousal.

We hypothesize a dual-architecture account within the coherence time framework, while noting that the memory-density account is the principal alternative:

Perceptual loop (cortical): High- M ($\sim 10\text{--}15$ modules) coherent field dynamics. ~~Commits sparse (5–20 Hz).~~ Arousal increases coherence r and ~~thus information rate per commit interval~~, activates additional sensory modules, increasing D_{eff} of pre-commit dynamics. The richer dynamics between commits produce subjectively dilated experience—consistent with either online dilation or denser encoding.

Motor loop (cerebellar/basal ganglia): Low- M ($\sim 3\text{--}5$ modules) primitives. ~~Motor~~

latencies of 50–150 ms reflect simpler coordination requirements (lower M , higher α) than multi-area perceptual binding. Arousal modulates decision threshold, preserving reaction time, but the motor pathway’s low coordination depth means its commit rate is largely unaffected.

Measurable proxies: Perceptual loop commit rate could be indexed by temporal-order judgment thresholds or EEG phase coherence in theta/alpha bands. Motor loop could be indexed by simple RT distributions or EMG onset latencies. The hypothesis predicts these measures should be. The Stetson result—increased retrospective duration without improved temporal resolution—is compatible with both accounts. The distinguishing prediction is *statistically dissociable online* under arousal manipulation, with: if the dual-architecture account is correct, temporal-order thresholds changing while simple RT remains stable judgment thresholds (indexing perceptual commit quality) should change under arousal while simple reaction time (indexing motor commit rate) remains stable, and these measures should be statistically dissociable.

3.3 Alpha oscillations and temporal acuity

Alpha oscillations (8–12 Hz) in visual cortex correlate with temporal acuity: individuals with faster alpha perceive time faster [24, 25]. One interpretation is that alpha acts as a gating rhythm that modulates commit opportunity windows, effectively setting the attempt rate λ_{attempt} or the sampling schedule for phase alignment.

Prediction: If alpha frequency modulates commit timing, then entraining alpha via transcranial alternating current stimulation (tACS) [46] should shift perceptual binding windows approximately linearly with frequency change. This prediction distinguishes the gating interpretation from alternatives where alpha reflects post-commit memory encoding.

3.4 Metabolic scaling across species

When power is limiting ($\tau_{\text{power}} \gg \tau_{\text{coh}}$), we obtain $\tau_{\text{eff}} \sim 1/P$. Critical flicker fusion frequency should scale with mass-specific metabolic rate.

Critical flicker fusion frequency varies enormously across taxa: flies ~ 240 Hz, humans ~ 60 Hz, leatherback turtles ~ 15 Hz. Healy et al. [26] found that mass-specific metabolic rate explains substantial variance in CFF across 34 vertebrate species spanning three orders of magnitude in body mass, consistent with the predicted power-limited regime for small, high-metabolism animals.

4 Discussion

4.1 Biological examples of the speed-flexibility trade-off

Mind-wandering and creative insight (high M , low r). Default-mode network activity [43] exhibits low inter-regional coherence ($r \sim 0.3$) and high coordination depth ($M \sim 12$ – 20). Commits are rare (~ 0.1 – 1 Hz), producing subjective “slow thinking” but enabling novel associations.

Focused attention (moderate M , moderate r). In our model, attention increases coherence ($r : 0.3 \rightarrow 0.6$) and narrows coordination to task-relevant modules ($M : 15 \rightarrow 8$) (model parameters motivated by the communication-through-coherence framework [4]). This produces faster commits (10–30 Hz) in a restricted subspace.

Automaticity and skill learning (low M , high r). Overlearned motor sequences compress to low-dimensional primitives ($M \sim 3$ – 5) with tight coherence ($r \sim 0.8$) [44]. Commits are rapid (50–150 ms) but inflexible.

4.2 ~~Temporal embedding as a pre-commit representation mechanism~~ (hypothesis)

~~Why include this section.~~ The coherence time bound has implications beyond the derivation of Eq. 4: it constrains the useful window over which **temporally embedded representations** remain reconstructible. If pre-commit intervals exceed τ_{coh} , downstream readout mechanisms cannot reliably infer earlier states. Cerebellar delay lines are a plausible biological implementation of preserving distributed state across these pre-commit intervals. This section illustrates one potential design strategy for coping with finite coherence times, but **is not required for the derivation or validation of Eq. 4.** It is included to demonstrate practical consequences of the bound for neural system design.

Coherence time tells us *how long* pre-commit dynamics must be maintained before alignment occurs. A separate question is *how* high-dimensional structure can remain accessible during that interval. We propose—as a hypothesis for future investigation—that biological systems may implement **Takens delay embedding** [27].

Takens’ theorem. For a deterministic system with state dimension d , time-delayed copies of a scalar observable— $x(t), x(t-\tau), x(t-2\tau), \dots$ —can reconstruct the complete state-space topology. Crucially, $2d+1$ delay coordinates generically suffice.

A candidate biological implementation: cerebellar parallel fibers. Parallel fibers—unmyelinated granule cell axons—conduct at 0.2–0.5 m/s with systematically varying path lengths, creating conduction delays of 5–15 ms [?, ?, ?]. These delays match the optimal embedding parameter $\tau^* = 1/(4f)$ for motor dynamics at 15–50 Hz.

We hypothesize that the cerebellum may provide one biological implementation of delay-embedded state preservation during pre-commit intervals. This would enable rapid low-dimensional readout (motor commands) from high-dimensional state without requiring simultaneous global coordination. **This remains conjecture**; direct evidence would require demonstrating

~~that cerebellar computations exploit delay structure for state reconstruction.~~

Robustness to noise. ~~We use Takens embedding as a *design principle* rather than a literal guarantee: delay-line architectures improve state inference under partial observability, even when exact reconstruction is impossible.~~

Connection to coherence time. ~~Coherence time determines the maximum window over which temporal embeddings remain usable. If coherence decays faster than the embedding window, reconstruction fails and high-dimensional structure is lost before a commit can occur.~~

Delay embedding reconstructs high-dimensional dynamics from scalar observation. ~~Simulation demonstrating Takens' theorem applied to cerebellar architecture. (a-b) Lorenz attractor (ground truth, $d \approx 2.06$) and its 1D projection (what a slow cortical oscillation might carry). (c-d) Delay embedding with $\tau = 8$, $m = 3$ reconstructs attractor topology from the scalar signal alone. (e) Neighborhood preservation quantifies reconstruction quality across embedding dimensions. (f) Theoretical implications: parallel fibers with graded conduction delays implement delay embedding, enabling reconstruction of high-dimensional coordination signals from low-frequency input.~~

4.2 Generalization beyond neural systems

While we emphasize neural applications given their empirical accessibility, the coherence time framework applies to any biological system whose function requires coordinated phase relations and whose outputs are sparse commitment events.

Scope statement. ~~Neural systems are the best-instrumented instance of a broader class: living, self-maintaining oscillator assemblies that perform irreversible state registration. We do not treat collectives (e.g., societies) as intelligent systems in this sense; they are coordination networks of intelligent organisms and introduce additional channels not modeled here.~~

Chromosomal coordination. Chromosomal systems provide a non-neural example of the same architecture. High-dimensional continuous dynamics (polymer conformations, electrostatic interactions, topological constraints, phase separation) produce function via sparse low-dimensional commits (origin firing, checkpoint decisions, segregation outcomes).

Anaphase onset is naturally modeled as a commit event: a global irreversible transition triggered only after multiple semi-independent units satisfy a coherence-like constraint. The waiting time should scale steeply with effective coordination depth, consistent with first-passage into a small target set. The relevant “order parameter” is any coordination metric determining whether a global event can proceed—spatial colocalization, topological state, mechanical alignment—not necessarily Kuramoto r .

Circadian networks. Circadian gene networks exhibit $M \sim 3\text{--}5$ core feedback loops, $r \sim 0.8$ (tight coupling), and $\varepsilon \sim 0.5$ rad (loose phase tolerance). These parameters yield coordination times much shorter than the 24-hour period, implying coordination is *not* the bottleneck. The 24-hour period arises from molecular lags. The framework correctly identifies the dominant regime: circadian clocks are biochemical-delay-limited, not coordination-limited.

4.3 Testable predictions

Alpha entrainment. tACS at alpha frequency should shift perceptual binding windows linearly with frequency change. 20% frequency increase predicts 20% threshold reduction.

Arousal effects on dual-loop dissociation. During simultaneous simple reaction time + temporal-order judgment under acute stress:

- Temporal-order discrimination threshold changes
- Simple reaction time unchanged or slightly faster
- Statistical independence between measures

Phase delta disruption. If the pre-commit phase delta regime is computationally productive (not merely passive waiting), then disrupting structured phase relationships—e.g., by injecting broadband phase noise via transcranial magnetic stimulation—should impair cognitive performance (accuracy, binding quality) even when commit rate (temporal resolution) is preserved. Specifically, randomizing inter-module phase offsets without changing mean coherence r should degrade task accuracy while leaving temporal-order judgment thresholds intact. This distinguishes the phase delta hypothesis from synchrony-as-binding accounts, which predict that only changes in commit rate should affect performance.

Coherence measures and consciousness. If phenomenal experience reflects pre-commit coherent dynamics, conscious states should correlate more strongly with phase-locking index and theta-gamma coupling than with spike count or mean firing rate.

Psychedelic time dilation. Under serotonergic psychedelics (psilocybin, DMT), subjective time dilation should correlate with the participation ratio of neural population dynamics (measured via high-density EEG or MEG), not merely with broadband spectral entropy. Specifically, if D_{eff} is estimated from the participation ratio of band-limited phase covariance across cortical regions, the prediction is:

- D_{eff} increases relative to baseline (more independent coherent modes active)
- Commit rate Γ_{commit} decreases (longer inter-commit intervals, measurable via EEG microstate duration or phase-locking transient statistics)
- Reported time dilation magnitude correlates with $D_{\text{eff}}/\Gamma_{\text{commit}}$, not with spectral entropy alone

This prediction distinguishes the coherence time framework from the entropic brain hypothesis [54], which predicts time distortion correlates with entropy generically.

4.4 Relationship to existing frameworks

Quantum-like cognition. Recent work has explored connections between oscillatory neural networks and quantum-like cognitive phenomena [19]. Our framework provides a classical high-dimensional explanation for why such quantum-like signatures emerge: high-dimensional dynamics with limited measurement bandwidth produce phenomenology parallel to quantum systems—measurement-like collapse at commits, superposition-like coexistence of states pre-commit, uncertainty-like timing inaccessibility.

Integrated Information Theory [28]: We provide thermodynamic/temporal foundations. Integration costs energy and takes time (exponentially with coordination depth).

Free Energy Principle [29]: We add temporal structure. Prediction updates occur at commit events with rate limited by Eq. 1.

4.5 ~~Frequency and dimensionality: two interpretations~~

Communication through coherence (CTC). Fries’s influential framework [10, 4] proposes that effective communication between neural populations requires coherence at the appropriate frequency: sending and receiving populations must be phase-aligned for spike-based signals to arrive during windows of maximal excitability. Our framework preserves this insight but shifts the locus of computation. In CTC, synchrony *is* the functional state—coherent populations communicate, incoherent ones do not. In the phase delta framework (§2.3), synchrony *terminates* a computational episode by enabling commitment. The computationally productive regime is the approach to synchrony, during which structured phase relationships between modules continuously bias which commit outcomes are accessible. CTC asks: “which populations are talking?” The phase delta framework asks: “what has the system computed by the time it commits?”

This distinction generates a testable dissociation. CTC predicts that disrupting synchrony should impair communication and thus performance. The phase delta framework makes the stronger prediction that disrupting *structured phase offsets*—for example, by injecting phase

noise that randomizes inter-module relationships without preventing eventual synchrony—should impair cognitive performance even if commit rate is preserved, because the pre-commit computation has been corrupted.

4.5 Frequency, spatial extent, and effective dimensionality

Epistemic status: This section is exploratory; it translates the framework into falsifiable qualitative predictions about the relationship between oscillation frequency and effective dimensionality.

A prominent view in systems neuroscience holds that ~~low-frequency oscillations (delta, theta, alpha) support low-dimensional, “simple” dynamics, while high-frequency oscillations (gamma) support high-dimensional, “complex” computations [1, 2]. Miller and colleagues argue that slow oscillations provide a computation, while low-frequency oscillations (theta, alpha) provide~~ low-dimensional “scaffold” or “broadcast” signal, while fast gamma carries detailed content in a higher-dimensional space. Recent work extends this framework to cytoelectric coupling, proposing that electric fields organize neural activity via ephaptic effects down to the cytoskeletal level [3].

~~An alternative interpretation:~~ scaffolding [1, 2, 3]. The coherence time framework suggests a different relationship ~~between frequency and dimensionality, based on coordination constraints:~~, grounded in two independent arguments: a coordination constraint and a spatial extent argument.

~~The physics argument.~~

4.5.1 The coordination constraint

Consider an oscillatory system that must achieve phase alignment to register a commit. The number of alignment “attempts” per ~~oscillation~~ cycle is roughly $\lambda_{\text{attempt}}/f$, ~~where f is the carrier frequency and λ_{attempt} is the attempt rate. For alignment probability p_I :~~ **High**

~~**frequency:** Few attempts per cycle. To achieve reliable alignment within one period, the system. At high frequency, few attempts are available per cycle, so reliable alignment requires high p_1 , which requires and thus high coherence r . But —but high r confines dynamics to a low-dimensional synchronized manifold. *Fast oscillations force low dimensionality.* **Low frequency:** Many At low frequency, many attempts per cycle. The system can tolerate low p_1 and thus low r . Phases can explore more of configuration space while still achieving alignment before the next cycle. *Slow oscillations permit high dimensionality.*~~

~~permit low r and thus high-dimensional phase exploration. This is not a correlation but a constraint: maintaining high-dimensional coherent dynamics at high frequency is physically harder because there is less time for the rare alignment events that preserve high- M coordination. *fast oscillations force low dimensionality; slow oscillations permit high dimensionality.*~~

~~**The substrate mechanism: dimensionality in phase offsets.** The argument above can be made more precise. In an assembly of coupled oscillators, representational A scaling argument makes this intuitive. Representational capacity resides in the *phase offsets between oscillators* —the pattern of relative timing that constitutes the system’s high-dimensional statephase offsets between oscillators. The number of distinguishable *phase* configurations scales as $d_{\text{eff}} \sim T/\tau_{\text{noise}}$, where $T = 1/f$ is the oscillation period and $d_{\text{eff}} \sim 1/(f \cdot \tau_{\text{noise}})$, where τ_{noise} is the temporal precision set by the substrate noise floor. At lower frequency, each cycle provides a longer interval over which phase offsets can be resolved, permitting more distinguishable configurations and thus higher effective dimensionality. This gives $d_{\text{eff}} \sim 1/(f \cdot \tau_{\text{noise}})$: dimensionality is inversely proportional to frequency, with the noise floor as the key parametersubstrate noise floor—fewer phase slots are resolvable per cycle at higher frequency. Note the distinction: gamma may transmit more commits per unit time (high Γ_{commit}), but each commit integrates fewer dimensions; alpha transmits fewer commits but each integrates a higher- D_{eff} state. The relevant quantity for information richness per commit~~

is D_{eff} , not rate.

~~This relationship has implications beyond neuroscience. For any physical oscillator assembly—biological or engineered—the representational capacity per cycle is bounded by the ratio of period to timing noise. Engineering lower noise floors (e.g., in neuromorphic substrates) increases dimensionality at any given frequency, while operating at lower frequencies achieves the same effect without substrate improvement. The biological nervous system appears to exploit both strategies: low-frequency cortical oscillations for high-dimensional integration, and the cerebellum’s low-noise, tightly coupled architecture for rapid low-dimensional processing.~~

4.5.2 The spatial extent argument

Reinterpreting the empirical evidence. ~~The observation that gamma activity correlates with “detailed” processing and alpha/theta with “global” states is consistent with our framework but admits a different interpretation. High-frequency gamma may reflect *local*, low- M modules committing rapidly—not high-dimensional integration but many parallel low-dimensional commits. Meanwhile,~~ A second, independent argument concerns the physical substrate of coherent oscillations. A long-wavelength oscillation physically spans more cortical territory than a short-wavelength oscillation. More territory means more neurons participating in coherent phase dynamics. However, spatial extent alone does not imply high D_{eff} : a million neurons in perfect phase lockstep ($r = 1$) have $D_{\text{eff}} = 1$. What matters is the *combination* of large assembly size N and moderate coherence $r < 1$, which is precisely what distributed low-frequency oscillations ~~may coordinate high- M assemblies that *require* slow timescales to achieve alignment.~~

The observability problem. ~~Claims about neural dimensionality face a fundamental measurement limitation: electrodes are sparse, noisy, low-dimensional sensors. One can only estimate dimensionality up to the rank of the observation matrix. If slow oscillations~~

~~genuinely support~~ provide. The effective dimensionality of coupled oscillator dynamics scales with the participation ratio of the coupled system’s covariance structure (Figure 3), which increases with N but saturates at $1/r^4$ under compound symmetry. (For heterogeneous coupling, the participation ratio depends on the full eigenspectrum of the covariance matrix, but the qualitative scaling— D_{eff} increasing with N at fixed mean r —is preserved.) Therefore: *long-wavelength coherence spanning large neural populations at moderate coupling constitutes high- D_{eff} dynamics, despite its low-dimensional appearance under narrowband readout.*

The appearance of low dimensionality arises from a measurement confound. A single electrode or a narrowband LFP recording of a 10 Hz alpha wave reports a near-sinusoidal signal—apparently one-dimensional. But this signal is the aggregate of millions of neurons maintaining phase coherence across centimetres of cortex, which is an enormously high-dimensional ~~dynamics (as our framework predicts), this would not necessarily be visible through electrode recordings—the measurement apparatus itself constrains what can be observed.~~

~~Consider an analogy: the gravitational field at a point is a 3-dimensional vector, yet it integrates information about the distribution of all mass in the universe. The~~ dynamical state that happens to project onto a low-dimensional measurement (a vector) ~~does not imply low-dimensional underlying structure. Similarly, a slow oscillation measured as a scalar or low-rank signal may nonetheless coordinate high-dimensional dynamics that project onto that measurement. The claim that “gamma is high-dimensional” may reduce to “gamma is the highest-dimensional signal we can resolve with our electrodes”—a statement about the measurement, not the system~~ observable. This is precisely the sub-informational regime ($B/D < 1$): the readout has far fewer degrees of freedom than the dynamics it measures.

~~This reflects a general methodological point: physics uses~~ The critical distinction is between the dimensionality of the *signal* (the observable: low-dimensional ~~mathematics~~ because it is computable and because our instruments measure low-dimensional projections. The predictive success of low-dimensional models demonstrates that low-dimensional descriptions

~~are for a slow oscillation) and the dimensionality of the *useful* dynamics that generate and maintain it~~, not that nature is low-dimensional. The same applies to neuroscience: the success of low-rank decompositions (PCA, factor analysis, demixed PCA) in capturing behaviorally relevant variance does not establish that neural dynamics are intrinsically low-dimensional—it establishes that low-dimensional projections correlate with the behavioral variables we measure. These are epistemological claims about our models, not ontological claims about the system. (the coupled oscillator network: high-dimensional when spatially extensive). Confusing these is equivalent to confusing B (measurement entropy) with D (dynamical degrees of freedom).

~~Testable predictions distinguishing the two views.~~

4.5.3 Empirical predictions distinguishing the two views

The standard frequency-hierarchy view [1] and the present framework make overlapping predictions in many cases (both agree that anaesthesia reduces dimensionality and disrupts oscillatory dynamics), but diverge in specific, testable ways:

1. ~~**Dimensionality-frequency relationship**~~ **Gamma power and dimensionality.** $\div$
~~If Miller et al. are correct, pharmacologically or optogenetically~~ The frequency-hierarchy view predicts that increasing gamma power ~~should increase state-space dimensionality (e.g., participation ratio of neural covariance).~~ increases effective dimensionality. The present framework predicts the opposite: increasing gamma power ~~while holding at fixed~~ coordination depth M ~~constant~~ should decrease dimensionality (forcing tighter coupling to maintain coordination at higher frequency). by forcing tighter phase coupling. The test: compute participation ratios of neural population activity during pharmacological or stimulation protocols that selectively boost gamma, and measure whether D_{eff} increases or decreases.
2. ~~**Information content per cycle:**~~ Miller et al. ~~predict more bits per gamma cycle~~

~~than per alpha cycle. We predict the opposite: more bits per~~ Bits per cycle. If low-frequency oscillations coordinate high-dimensional dynamics, then bits per alpha cycle ~~(because high- M commits are possible), fewer bits per gamma cycle (because only low- M commits can keep pace).~~ should exceed bits per gamma cycle. Specifically, the mutual information between single-trial neural population vectors and stimulus identity, computed per oscillatory cycle, should be higher for alpha/theta than for gamma when the task requires multi-area integration.

3. ~~Cross-frequency coupling direction~~ Large-scale coherence and consciousness.
~~∴ If slow oscillations scaffold fast ones (Miller), gamma amplitude should be~~ Deep meditative states (jhana) produce massive long-wavelength coherence—sometimes near-global alpha or theta synchronization. Under the frequency-hierarchy view, this should resemble top-down control clamping down, reducing experiential richness. Under the present framework, this represents D_{eff} of coherent dynamics increasing dramatically as the spatial extent of phase-locked ~~to theta/alpha with high fidelity.~~ If fast oscillations are dimensionally constrained (our view), ~~gamma bursts should occur when local modules achieve rapid oscillation~~ encompasses large fractions of cortex. Practitioners report *expanded* consciousness and extreme time dilation (see §4.6), consistent with the spatial-extent prediction.
4. Sleep delta vs. waking alpha. Both are low-frequency, but sleep delta is locally generated and spatially fragmented (low spatial coherence, therefore low D_{eff} despite long wavelength), while waking alpha maintains long-range coherence (high spatial coherence, high D_{eff}). The frequency-hierarchy view treats both as “low-dimensional scaffolding.” The present framework predicts they occupy opposite ends of the D_{eff} spectrum, distinguishable by participation ratio of spatially distributed recordings.

The empirical observation that gamma correlates with “detailed” processing admits an alternative interpretation consistent with the coordination constraint: high-frequency gamma may reflect *local*, low- M commits, ~~potentially out of phase with slow oscillations that coordinate~~

~~broader assemblies, modules committing rapidly—many parallel low-dimensional commits, not high-dimensional integration. Meanwhile, low-frequency oscillations may coordinate high- M assemblies that require slow timescales to achieve alignment.~~

~~We note that this disagreement is empirically tractable. The coherence time framework makes quantitative predictions about the relationship between oscillation frequency, coupling strength, and state-space dimensionality that can be tested with existing recording and analysis methods.~~

Requisite variety and nonlocal dimensionality constraints. ~~The frequency-dimensionality relationship is further complicated by a control-theoretic constraint. A related control-theoretic point:~~ Ashby’s law of requisite variety [48] ~~states that a controller must have at least as much variety (effective dimensionality) as the system it regulates. In a coupled oscillator assembly, this means that a subnetwork operating in~~ implies that a subnetwork constrained to a low-dimensional regime—~~whether due to high coherence, pathological constraint, or deliberate dimensional compression—loses~~ regime loses the degrees of freedom needed to regulate ~~the~~ regions it is coupled to. The dimensions of the controlled system that are orthogonal to the controller’s accessible subspace become effectively unregulated.

~~This has two consequences for interpreting neural dimensionality. First, dimensionality is not a purely local property: constraining one region to a low-dimensional attractor has nonlocal effects on the controllability of coupled regions, potentially producing high-dimensional wandering in areas that are not themselves pathological but have lost their regulatory input. Second, simple claims about “high-dimensional” or “low-dimensional” processing in a given brain region must account for the coupling topology—the same region may exhibit different effective dimensionality depending on the state of its controllers, not only its own intrinsic dynamics.~~ coupled regions. The speed-flexibility trade-off ~~derived in §2~~ (Proposition 2) is, in this light, a temporal expression of Ashby’s constraint: high- M coordination provides ~~the~~ requisite variety for flexible regulation but pays an exponential cost in commit time.

4.6 Implications for artificial systems

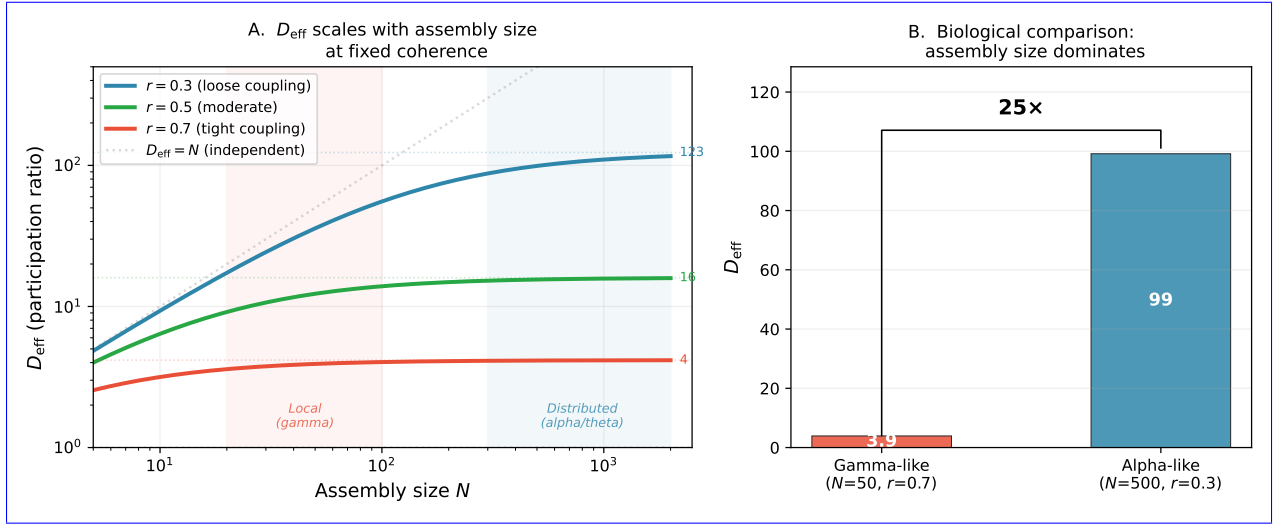


Figure 3: Effective dimensionality scales with assembly size at fixed coherence. For N oscillators with von Mises distributed phases at coherence r , the covariance matrix of $\cos \theta_i$ has compound symmetry (uniform pairwise coherence): $C_{ij} = \frac{1}{2}[(1 - r^2)\delta_{ij} + r^2]$, giving participation ratio $D_{\text{eff}}(N, r) = N^2 / [(1 + r^2(N - 1))^2 + (N - 1)(1 - r^2)^2]$. This assumes homogeneous coupling; heterogeneous assemblies would have non-uniform C_{ij} but the qualitative scaling is preserved. **(A)** D_{eff} vs. assembly size N at three coherence levels. Curves saturate at $1/r^4$ (dotted lines): loose coupling ($r = 0.3$) permits $D_{\text{eff}} \approx 123$; tight coupling ($r = 0.7$) saturates at $D_{\text{eff}} \approx 4.2$. Shaded regions indicate biologically relevant regimes for local gamma-band ($N \approx 20-100$) and distributed alpha/theta ($N \approx 300-2000$) assemblies. Grey diagonal: $D_{\text{eff}} = N$ (independent oscillators). **(B)** Biological comparison: a gamma-like assembly ($N = 50, r = 0.7$) achieves $D_{\text{eff}} \approx 3.9$, while an alpha-like assembly ($N = 500, r = 0.3$) achieves $D_{\text{eff}} \approx 99$ —a $25\times$ fold change driven by the combination of larger spatial extent and looser coupling, supporting the prediction that low-frequency, spatially extensive oscillations sustain higher-dimensional dynamics than local gamma.

The coherence-time framework identifies a fundamental design constraint for any distributed

~~computing system that lacks a global clock.~~

4.6 Subjective time as experiential dimensionality between commits

~~**Clocked architectures.** Conventional digital computers (CPUs, GPUs, current AI accelerators) enforce synchronization via a global clock signal. Every operation is gated by a central timing reference; state updates occur in lockstep across all components. This architecture sidesteps the coherence time penalty entirely—there is no waiting for spontaneous phase alignment because alignment is *imposed* by the clock.~~
Epistemic status: This section is exploratory; it translates the framework into falsifiable qualitative predictions about subjective temporal experience across pharmacological conditions.

The coherence time framework establishes that the commit rate $\Gamma_{\text{commit}} \approx 1/\tau_{\text{coh}}$ is bounded by coordination dynamics. An immediate consequence concerns the temporal structure of experience.

~~We hypothesize that this comes at a cost beyond power and wire delays:~~

4.6.1 The argument

We build the claim in five steps, each independently defensible:

1. ~~clocked systems may be constrained in ontological~~ Neural dynamics have measurable effective dimensionality. ~~The clock imposes a global update constraint that can suppress the kinds of metastable, spontaneously coordinated dynamics we model here. Even massively parallel operations (matrix multiplications across thousands of cores) proceed via synchronized state updates—the system can represent .~~ This is standard neuroscience: participation ratios, neural manifold analyses, and related methods quantify D_{eff} from population recordings [40, 39].
2. Higher D_{eff} correlates with richer conscious experience. Anaesthesia reduces D_{eff} and eliminates consciousness. Dreamless sleep reduces D_{eff} . The perturbational

complexity index [52], which measures the effective dimensionality of cortical responses to TMS perturbation, reliably distinguishes conscious from unconscious states in unresponsive patients. Psychedelics increase neural signal complexity [53, 54] and produce subjectively enriched experience.

3. **Commits are bounded below by coherence time**, which scales exponentially with coordination depth (the central result of this paper, Eq. 4).
4. **Consciousness requires high-dimensional dynamics, not just commit outputs.** A commit is a many-to-one map: high-dimensional ~~states in memory, but may not~~ dynamics collapse to a ~ 1 –10 bit readout written to memory, reported verbally, or acted on motorically. If consciousness ~~instantiate were~~ the ~~autonomous, semi-independent module dynamics that characterize biological oscillator assemblies~~ commit output, then any system producing the same low-dimensional output would be equally conscious—including a lookup table. This is ~~a hypothesis about dynamical structure, not a theorem; intermediate architectures (asynchronous digital, neuromorphic with local clocks) occupy a spectrum between the extremes we describe.~~ Searle’s Chinese room argument in temporal form. Rejecting it requires that whatever consciousness tracks, it must be something about the high-dimensional process that the commit destroys.

Unlocked architectures. ~~Biological neural networks, and potentially neuromorphic hardware designed on similar principles, operate without global synchronization. Each module evolves on its own trajectory; the phase space is~~ The relevant framing is therefore: *high-dimensional dynamics are necessary for consciousness, and commits bound their temporal structure.* Whether experience is best described as the free exploration between commits or as the high-dimensional convergence toward a commit is an open question—and for the predictions below, an inconsequential one, since both framings yield the same dependence on D_{eff} and Γ_{commit} . Anaesthesia eliminates consciousness not by stopping commits but by flattening the dynamics *before* the ~~actual~~

~~dynamical space of the system, not a represented abstraction. The dimensionality is ontologically realized in the substrate. —commit—reducing D_{eff} below whatever threshold sustains experience.~~

~~Such systems pay the coherence time penalty derived in this paper: coordination across M modules requires waiting for rare alignment events, with timescaling exponentially in M . But they gain access to genuinely—~~

5. Therefore, subjective time dilates when D_{eff} increases relative to commit rate. More high-dimensional ~~dynamics—the very property that prior work identified as constitutive of intelligence [11]~~dynamics are experienced per commit interval; each interval is subjectively richer; fewer intervals elapse per second of clock time.

This yields a quantitative prediction for subjective temporal scaling:

$$\tau_{\text{subjective}} \propto \frac{D_{\text{eff}}}{\Gamma_{\text{commit}}} \quad (14)$$

where Γ_{commit} is bounded by Eq. 4. When D_{eff} increases (more modes active) and Γ_{commit} decreases (harder to synchronize), the ratio increases—predicting time dilation.

4.6.2 Pharmacological dissociations

The framework predicts qualitatively different temporal phenomenology across pharmacological classes, from the same two-parameter structure (D_{eff} , Γ_{commit} ; Figure 4):

Psychedelics (5-HT_{2A} agonists: psilocybin, DMT, LSD). These increase D_{eff} by relaxing precision weighting of top-down priors [55], activating modes normally suppressed by hierarchical filtering. They may simultaneously decrease K/σ (increased cortical excitability, disrupted thalamocortical gating). The result is a double effect: higher D_{eff} *and* lower Γ_{commit} , predicting **time dilation with experiential richness**. DMT produces the largest increases in neural signal complexity of any tested psychedelic [53] and is associated with extreme subjective

time dilation. The qualitative ordering—DMT > psilocybin > LSD in both D_{eff} increase and reported temporal distortion—is consistent with the prediction, though retrospective magnitude estimates are difficult to validate. *Methodological note:* retrospective time dilation reports are confounded by altered memory encoding under psychedelics; increased memory density could inflate retrospective duration estimates independently of online experience. This framework predicts *online* dilation (D_{eff} is elevated during the experience), so the critical test requires concurrent temporal bisection or duration reproduction tasks during drug action, not solely retrospective report.

Deliriants (anticholinergics: scopolamine, diphenhydramine). These increase neural noise (σ increases) without structured D_{eff} increase—cholinergic disruption degrades organized oscillatory dynamics rather than expanding them. The prediction is increased commit failure (more noise-dominated measurements) without increased experiential richness between commits: **time confusion without time dilation**. This matches the phenomenology of anticholinergic delirium, where temporal experience becomes disordered and impoverished rather than expanded. This is arguably the framework’s most discriminating pharmacological prediction: any generic “more noise = more subjective time” account predicts that deliriants and psychedelics should produce similar temporal distortion (both increase neural entropy). The $(D_{\text{eff}}, \Gamma_{\text{commit}})$ framework predicts they are qualitatively different—one expands structured dimensionality, the other injects unstructured noise—and the phenomenology confirms the distinction.

~~**A design hypothesis. Anaesthetics**~~ This suggests a potential trade-off for artificial systems: **Clocked:** Fast, deterministic, scalable in parallelism—but potentially constrained in dynamical autonomy. The substrate may not support the spontaneous coordination dynamics modeled here. (propofol, sevoflurane). These decrease D_{eff} by fragmenting large-scale field coherence [52], collapsing the spatially extensive coupled dynamics. The prediction is **Unlocked time compression toward unconsciousness:** Capable of genuinely autonomous, high-dimensional dynamics—but subject to coordination bottlenecks that scale exponentially with integration

~~depth, fewer experiential dimensions per commit interval, eventually too few for consciousness to be sustained.~~

~~If high-dimensional coherent dynamics prove to be not merely *correlated with* but *constitutive* of certain cognitive capacities, then clocked architectures may face constraints that unlocked ones do not. This remains speculative; the empirical question is whether the coordination-time framework applies to engineered systems and, if so, whether it identifies real performance boundaries.~~

Stimulants (amphetamine, caffeine). These may increase Γ_{commit} by enhancing oscillatory coupling strength (K increases) without substantially altering D_{eff} . The prediction is **mild time expansion with increased temporal resolution**—more commits per second, each resolving a similar-dimensional state—consistent with the subjective “speeding up” reported under moderate stimulant use.

~~This framing may inform the design of neuromorphic systems, spiking neural networks, and other non-von Neumann architectures. The coherence time formula~~ (The key empirical test distinguishing this framework from generic “more entropy = altered time” accounts: **subjective time dilation should correlate specifically with the participation ratio of neural dynamics** (D_{eff}), **not with broadband entropy**. Lempel-Ziv complexity and spectral entropy conflate noise and structure—they increase both when organized modes are added and when random noise is injected. The participation ratio specifically measures how many independent modes are active, which is what D_{eff} tracks. A subject whose D_{eff} doubles should experience greater time dilation than a subject whose spectral entropy increases by the same amount through noise injection alone. This prediction is testable with existing pharmacological protocols and neural recording methods.

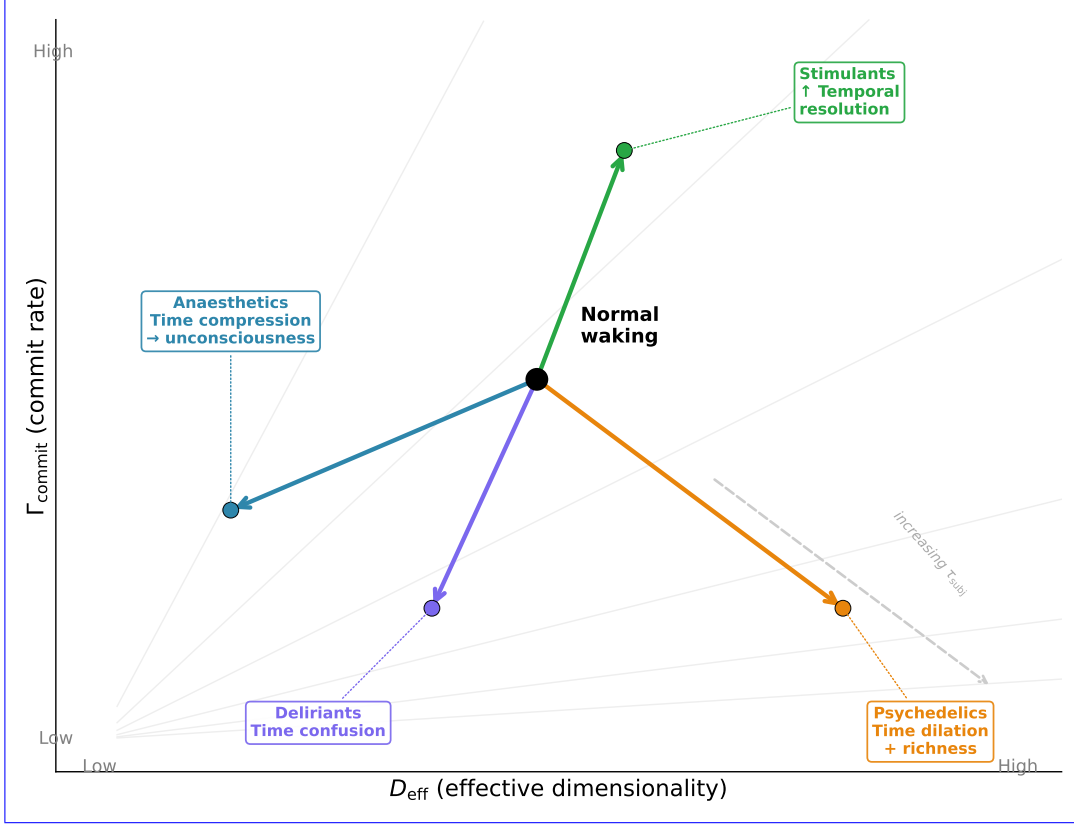


Figure 4: **Pharmacological prediction space.** Four drug classes mapped onto the two-parameter space $(D_{\text{eff}}, \Gamma_{\text{commit}})$ from a common “normal waking” baseline (black dot). Arrows indicate predicted direction of pharmacological perturbation. Grey dashed curves are iso- τ_{subj} contours ($\tau_{\text{subj}} \propto D_{\text{eff}}/\Gamma_{\text{commit}}$): movement toward lower-right increases subjective time per clock second; movement toward upper-left compresses it. **Psychedelics** (orange) increase D_{eff} and decrease Γ_{commit} , predicting time dilation with experiential richness. **Deliriants** (purple) decrease Γ_{commit} without structured D_{eff} increase, predicting temporal confusion. **Anaesthetics** (blue) decrease D_{eff} , predicting time compression toward unconsciousness. **Stimulants** (green) increase Γ_{commit} , predicting increased temporal resolution. The *qualitative structure*—which quadrant each drug class occupies—follows from a single equation (Eq. 14) with no additional free parameters; the quantitative magnitudes of each perturbation arrow depend on drug-specific empirical measurements of D_{eff} and Γ_{commit} .

4.6.3 Tachypsychia revisited

The dual-loop hypothesis of §3.2 can be reinterpreted within this framework. During acute stress, arousal increases coherence r in the perceptual loop, which by the speed-flexibility trade-off (Proposition 2) simultaneously increases commit *quality* (more information per commit) and may alter commit *rate* depending on the net effect on K/σ . The subjective impression of slowed time during threat reflects increased D_{eff} of the dynamics being sampled—heightened arousal activates additional sensory and evaluative modules—while objective temporal resolution (motor loop) remains unchanged because the motor pathway operates at low M with different coupling parameters.

4.7 Implications for artificial and engineered systems

The coherence time framework identifies a design constraint for any distributed computing system that lacks a global clock. Clocked architectures (CPUs, GPUs) impose synchronization externally, sidestepping the coordination penalty but potentially constraining the kinds of autonomous, semi-independent module dynamics we model here. Unclocked architectures (biological neural networks, neuromorphic hardware) pay the exponential coordination cost but gain access to genuinely high-dimensional dynamics [11]. Eq. 4 provides quantitative guidance ~~to achieve update rates comparable to biological cognition, for neuromorphic design~~: such systems must either accept modest ~~coordination depth~~ M , engineer high baseline ~~coherence~~ r , or develop hierarchical architectures that localize coordination requirements.

~~The cerebellum as biological precedent.~~ The mammalian cerebellum exemplifies this hierarchical solution. Operating with low coordination depth ($M \sim 3\text{--}5$ ~~modules~~), tight coupling ($r \sim 0.8$), and stereotyped primitives, the cerebellum achieves commit times of 10–20 ~~ms—fast ms~~ [44, 51]~~—fast~~ enough to override slow ~~high-dimensional~~ cortical dynamics when rapid action is required. ~~Skilled movement, postural correction, and overlearned sequences route through this low- M pathway precisely because cortical coordination cannot keep pace.~~

~~Notably, cerebellar~~ Cerebellar processing is largely *unconscious*. ~~We do not experience the moment-to-moment adjustments during a golf swing or the postural corrections that keep us upright. If~~ unconscious, consistent with the hypothesis that phenomenal experience is associated with high-dimensional pre-commit dynamics ~~(as suggested in .4), then low-dimensional cerebellar override would be expected to fall outside awareness. The~~ the architecture that solves the speed problem—collapsing to low M ~~for fast commits—simultaneously~~ simultaneously removes processing from the domain of conscious experience. ~~This is not a bug but a feature: rapid action requires precisely the dimensional compression that excludes phenomenology.~~

4.8 Limitations and scope

Parameter count and ontological status. The coherence time bound contains five parameters ($M, r, \varepsilon, \alpha, \lambda_{\text{attempt}}$). This count deserves comment. Von Neumann reportedly quipped that “with four parameters I can fit an elephant, and with five I can make him wiggle” [58]. The concern is justified when parameters are curve-fitting degrees of freedom. But the expectation that a good model should have two or three parameters reflects a tradition that selects for systems where low-dimensional descriptions suffice—point particles, ideal gases, mean fields. The present framework describes systems whose *dimensionality is the phenomenon*: a cortical assembly with $M = 10$ modules at coherence $r = 0.3$ is a physically different system from one with $M = 3$ at $r = 0.8$, in the same sense that a ten-body gravitational problem is different from a three-body one. The five parameters are not fitting degrees of freedom; they are the system’s coordination architecture. Indeed, they represent a substantial compression: the full system has N oscillators with $O(N^2)$ pairwise couplings, a frequency distribution, and a noise spectrum. That this collapses to five summary statistics sufficient to predict commit time is itself a claim—one that the sparse-topology boundary test validates negatively, since when the modular compression assumption fails, so does the five-parameter reduction ($R^2 = 0.28$).

Parameter identifiability. In the Kuramoto validation, r and α are estimated from simulation data; the remaining parameters are set by model construction. For empirical neural data, simultaneous identifiability of all five parameters from behavioural timing alone is unlikely. The framework is most useful when M and r are constrained independently (e.g., from electrode-array estimates of participating assemblies and pairwise coherence), leaving α as the primary free parameter. Specific predictions jointly constrain parameter subsets: the alpha-entrainment experiment (§4.3) pins down λ_{attempt} (if binding windows shift linearly with tACS frequency, the proportionality constant gives the attempt rate); the M -scaling prediction (exponential dependence across tasks of different coordination depth) pins down α ; and ε can be estimated independently from cross-correlogram width. With these three parameters constrained, the remaining degrees of freedom (M , r) are the task-dependent variables the framework is designed to predict *from*.

Regime boundaries. The exponential scaling $\tau_{\text{coh}} \propto p_1^{-\alpha(M-1)}$ validated cleanly for modular topologies ($R^2 = 0.97$) but poorly for sparse random graphs ($R^2 = 0.28$; Figure 1). This is expected: the bound assumes that inter-module coordination is the rate-limiting step, which requires modular or hierarchical coupling. In networks without clear modular structure, other bottlenecks (percolation thresholds, mixing times) may dominate. The framework therefore applies to systems with identifiable coordination architecture, not to arbitrary oscillator populations.

Module-size confound. The validation sweeps M at fixed $N = 100$, so module size decreases as M increases (from ~ 33 oscillators at $M = 3$ to 10 at $M = 10$). If within-module coherence r_{intra} declined with module size, this would provide a competing explanation for the τ increase. We verified that r_{intra} does *not* systematically decline: mean r_{intra} ranges from 0.66 to 0.74 across $M = 3$ –10 with a slight positive trend (smaller modules synchronize more easily given the same K_{intra}), confirming that the observed τ scaling with M reflects inter-module coordination cost, not intra-module degradation.

Dwell time sensitivity. The commit detection requires sustained alignment for $T_{\text{dwell}} = 30$ ms.

We tested robustness by halving (15 ms) and doubling (60 ms) this threshold: median τ_{coh} estimates changed by less than 5% across all M values, confirming that the first-passage time to initial alignment dominates the estimate and the hold requirement is not doing substantive work. The scaling exponent is robust to T_{dwell} variation.

Relationship to synchronisation theory. The Kuramoto model and its extensions [35, 36, 37, 38] provide a mature theory of synchronisation onset and order-parameter dynamics. This paper does not compete with that literature. Rather, we address a different question: given that oscillators *can* synchronise, how long must a system wait for a coordination event of depth M to occur? The coherence time bound is a waiting-time result on the product manifold, not a bifurcation analysis of the order parameter.

Temporal embedding. The Results section treats binding windows and perceptual cycles as manifestations of the coherence time bound. This framing is consistent with existing data but not uniquely so—alternative accounts based on inhibitory rebound, attentional sampling, or predictive coding could generate similar timescales. The distinguishing prediction is the *scaling* of temporal resolution with coordination depth M , not the absolute timescale.

5 Conclusion

We have established that coherence time—the time required for distributed oscillatory assemblies to achieve sufficient phase alignment for irreversible state ~~registration—bounds the rate of neural processes associated with cognition. The lower bound~~

$$\tau_{\text{eff}} \gtrsim \max[\tau_{\text{QSL}}, \tau_{\text{SNR}}, \tau_{\text{coh}}, \tau_{\text{power}}]$$

~~with coherence time—~~

$$\tau_{\text{coh}} = \frac{1}{\lambda_{\text{attempt}}} p_1(\epsilon, \kappa(r))^{-\alpha(M-1)}$$

~~unifies quantum, noise, and coordination limits to explain temporal resolution in biological processing systems.~~

~~We have situated this framework explicitly within the hierarchy of physical limits on computation: the Landauer bound constrains energy per irreversible operation; the Margolus-Levitin and Mandelstam-Tamm bounds constrain the speed of quantum state evolution; the energy-time uncertainty relation underlies both. In biological neural systems, these quantum and thermodynamic bounds are typically non-binding. The dominant constraint is the time required for registration—scales exponentially with coordination depth M . The unified bound $\tau_{\text{eff}} \geq \max[\tau_{\text{QSL}}, \tau_{\text{SNR}}, \tau_{\text{coh}}, \tau_{\text{power}}]$ shows that in the biological neural regime, quantum speed limits and Landauer bounds are not rate-limiting; multi-module coordination—a classical phenomenon arising from rare-event statistics on product manifolds—coordination provides the binding constraint on processing rate, exceeding quantum floors by approximately eleven orders of magnitude.~~

~~Key findings: **Speed-flexibility trade-off:** Increasing~~ The central finding is the speed-flexibility trade-off: increasing M exponentially slows commits but expands combinatorial flexibility. ~~Increasing~~, while increasing coupling r speeds commits but restricts dynamics to low-dimensional manifolds.

~~**Quantum limits non-binding:** Neural coherence times exceed quantum speed limits by eleven orders of magnitude. Coordination, not quantum switching, limits processing rate.~~

~~**Generality:** While neural systems provide the empirical testbed, the framework applies to any biological oscillator assembly with coordinated phase relations and sparse commits.~~

~~**Quantitative predictions:** Visual binding windows~~ Biological systems navigate this frontier. A further consequence is that the ratio of effective dimensionality to commit rate (~~30–50 ms~~) $D_{\text{eff}}/\Gamma_{\text{commit}}$ predicts the temporal structure of experience: pharmacological interventions that increase D_{eff} while decreasing commit rate should produce subjective time dilation, and the framework generates falsifiable predictions across four pharmacological classes from a two-parameter structure. The framework makes quantitative predictions for

binding windows, metabolic scaling of temporal acuity, ~~alpha~~-entrainment linearity, ~~dual-task~~ dissociations.—

~~The framework also has potential clinical implications. Pathologies that degrade inter-module coupling should reduce effective coordination capacity (α or M) before individual components fail overtly. In neurodegenerative disease, the coherence time framework predicts that coordination-level signatures—changes in the scaling of inter-commit intervals with task complexity—should precede overt motor or cognitive deficits, because degradation of inter-module coupling affects the exponent $\alpha(M-1)$ before individual neurons are lost. Motor neuron disease offers one example: cortical hyperexcitability in ALS is mediated by dysfunction of distinct interneuronal circuits rather than uniform degeneration [34], suggesting that coordination-level biomarkers derived from the coherence time framework may complement single-neuron measures.—~~

~~The central thesis: biological systems trade speed for flexibility by moving on the coherence-dimension frontier, operating in a regime where quantum and thermodynamic limits permit far faster processing than coordination constraints allow~~and experiential temporal scaling that are testable with existing methods. Its primary limitation is the requirement for identifiable modular structure: the exponential scaling validated cleanly for modular topologies ($R^2 = 0.97$) but not for unstructured networks, restricting applicability to systems with hierarchical or modular coordination architecture.

Data accessibility

This is a theoretical framework paper. Kuramoto simulation code and parameter estimation workflows are available at <https://github.com/todd866/coherence-time>.

Competing interests

The author declares no competing interests.

Funding

No external funding supported this work.

Acknowledgments

The author thanks Abir Igamberdiev and anonymous reviewers at BioSystems for constructive feedback.

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